
Biologically Plausible Heavy-Tailed Connectivity Enhances Generalization on Cognitive Tasks in Recurrent Neural Networks

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Abstract

While heavy-tailed synaptic weight distributions are pervasive in biological neural networks, their computational role—particularly in relation to generalization—remains poorly understood. To address this, we develop a novel optimal-transport-based optimization algorithm that incorporates key biological constraints, including Dale’s principle and heavy-tailed synaptic statistics, to train recurrent neural networks (RNNs) on a wide range of cognitive tasks. We show that these biologically constrained, heavy-tailed RNNs exhibit substantially improved generalization, which we further characterize within a PAC-Bayes framework. Our theoretical analysis and numerical experiments reveal two complementary mechanisms underlying this generalization enhancement. Topologically, heavy-tailed connectivity induces an effectively low-rank structure, which in turn yields low-dimensional neural dynamics. Geometrically, heavy-tailed connectivity intrinsically shapes task variable representations to lie near a linear manifold, thereby improving generalization for a linear readout strategy. Together, these results identify heavy-tailed connectivity as a biologically grounded intrinsic mechanism that promotes low-rank structure and favorable representational geometry, leading to improved generalization in flexible cognitive tasks.

1. Introduction

Heavy-tailed synaptic connectivity is a pervasive structural feature of biological neural circuits, consistently observed across the mammalian cortex (Lefort et al., 2009; Loewenstein et al., 2011), hippocampus (Ikegaya et al., 2013), and even the *Drosophila* central brain (Scheffer et al., 2020). Interestingly, similar weight statistics often emerge in artificial neural networks trained via stochastic gradient descent (Hodgkinson & Mahoney, 2021; Gurbuzbalaban et al., 2021; Jiao & Keller-Ressel, 2024), suggesting a potential fundamental principle shared between biological and artificial learning systems. However, despite its ubiquity, the precise computational role of heavy-tailed connectivity remains largely unknown. In both cognitive neuroscience and machine learning, a central goal is understanding generalization—the ability to apply learned knowledge to novel, unseen situations (Lake et al., 2015; Saxe et al., 2021). While the statistical properties of heavy-tailed distributions have been studied in isolation (Mahoney & Martin, 2019; Simsekli et al., 2020), a critical question remains: does this specific connectivity structure confer a distinct advantage for generalization in flexible cognitive tasks?

To this end, generally one must understand how structural statistics translate into functional dynamics. Recent theoretical advances have established a strong link between connectivity structure and emergent population dynamics (Iyer et al., 2013; Gerstner et al., 2014). In particular, seminal works on low-rank RNNs (Mastrogiuseppe & Ostojic, 2018) have demonstrated that constraining the rank of the connectivity matrix drives the network dynamics into low-dimensional subspaces, which are sufficient to support flexible cognitive computations, such as flexible timing tasks (Beiran et al., 2023). While these studies provide a powerful framework connecting structure (low-rank) to function (low-dimensional dynamics), they typically rely on explicitly constructed or idealized connectivity profiles. It remains unclear how biologically realistic heavy-tailed statistics shape those emergent neural dynamics. Specifically, the mechanism by which heavy-tailed connectivity might spontaneously induce the advantageous dynamical regimes required for robust generalization has yet to be fully elucidated.

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Systematically investigating this relationship requires a way to train RNNs that perform complex cognitive tasks while strictly adhering to specific weight distributions. While an approach incorporating distributional constraints into neural network training has gained traction (Zhong et al., 2019), existing methods are primarily limited to feedforward architectures or generative modeling. To our knowledge, there is a lack of general-purpose optimization frameworks capable of training RNNs on cognitive tasks while simultaneously enforcing specific, biologically plausible connectivity statistics. This methodological gap hinders the causal study of how weight distribution shapes cognitive function.

In this work, we bridge this gap and articulate the computational benefits of heavy-tailed connectivity. Our contributions are summarized as follows:

- We propose DSco-SGD, a novel optimization algorithm that enables the training of RNNs to perform cognitive tasks while satisfying strict distributional constraints (e.g., heavy-tailed weights) and Dale’s principle.
- Through extensive numerical experiments and theoretical analysis using the PAC-Bayesian framework, we demonstrate that heavy-tailed RNNs exhibit significantly better out-of-distribution generalization, particularly in tasks requiring flexible timing and working memory.
- We uncover two potential mechanisms underlying this performance: topologically, heavy-tailed statistics induce an effective low-rank structure that compresses neural trajectories into low-dimensional subspaces; geometrically, this structure facilitates the formation of linear manifold representations of task variables, which naturally support generalization to unseen task parameters.

2. Train RNNs with heavy-tailed connectivity

We investigate RNNs consisting of N units designed to perform cognitive tasks. The dynamics of the total input current x_i to the i -th unit are governed by

$$\tau \dot{x}_i(t) = -x_i(t) + \sum_{j=1}^N J_{ij} \varphi(x_j(t)) + \sum_{k=1}^{N_{\text{in}}} I_{ik} u_k + \xi_i, \quad (1)$$

where $\varphi(\cdot) = \tanh(\cdot)$ is the single neuron transfer function, J_{ij} is the strength of the connection from unit j to unit i , and ξ_i represents an independent Gaussian white noise process.

The i -th unit receives N_{in} task-dependent scalar inputs u_k for $k = 1, \dots, N_{\text{in}}$, each along a set of feedforward weights that define an input vector $I_i = \{I_{ik}\}_{k=1}^{N_{\text{in}}}$. Each readout of

the network is defined as

$$z_l(t) = \mathbf{w}_l^\top \varphi(\mathbf{x}(t)) = \sum_{i=1}^N w_{li} \varphi(x_i(t))$$

for l in $\{1, \dots, N_{\text{out}}\}$, which is a linear combination of the firing rates of all units along the output vector $\mathbf{w}_l = \{w_{li}\}_{i=1, \dots, N}$. The task determines the numbers N_{in} and N_{out} of input and output vectors.

Furthermore, the $N \times N$ matrix $J = (J_{ij})$ is constrained to satisfy the following biologically plausible features

- (C1) Non-reciprocal interactions (i.e. $J_{ij} \neq J_{ji}$ for some pairs (i, j)) and no self-coupling (i.e. $J_{ii} = 0$ for $\forall i$).
- (C2) Dale’s principle. The connectivity J is partitioned into Nf excitatory columns J^E and $N(1 - f)$ inhibitory columns J^I . Here, f is the proportion of excitatory neurons, and $1 - f$ is that of inhibitory neurons.
- (C3) Heavy-tailed connectivity. The non-zero weights J_{ij}^m are drawn from heavy-tailed distributions ρ^m . Here, $m = E, I$.

To train RNNs on cognitive tasks, we formulate an empirical risk (loss function) as

$$\mathcal{L}_S(J) = \frac{1}{N_{\text{tr}}} \sum_{q=1}^{N_{\text{tr}}} l(J, u^{(q)}, \hat{z}^{(q)}),$$

where $S = \{u^{(q)}, \hat{z}^{(q)}\}_{q=1, \dots, N_{\text{tr}}}$ is a dataset specific to a given task and

$$l(J, u^{(q)}, \hat{z}^{(q)}) = \frac{1}{N_{\text{out}}} \sum_{l=1}^{N_{\text{out}}} \sum_{t_1^{(q)} < t < t_2^{(q)}} \left| z_l^{(q)}(t) - \hat{z}_l^{(q)}(t) \right|^2.$$

This loss function quantifies the squared difference between the readout $z_l^{(q)}$ and the target output $\hat{z}_l^{(q)}$ for all trials $q = 1, \dots, N_{\text{tr}}$. Here, $t_1^{(q)}$ and $t_2^{(q)}$ correspond to the time boundaries taken into account for computing the loss function. We quantified the generalization error by

$$\varepsilon = \hat{\mathcal{L}}(J) - \mathcal{L}(J)$$

where $\hat{\mathcal{L}}(J) = \mathbb{E}[l(J, u, \hat{z})]$ is the expected risk.

We propose an asymptotic Distribution-Sign constrained SGD (DSco-SGD) method to train heavy-tailed RNNs by minimizing the loss function. As outlined in Algorithm 1, the proposed method comprises two distinct components, the first utilizes stochastic gradient descent (SGD)¹ for weight updates; the second leverages the theory of optimal transport², to adaptively transport weights to a new value satisfying some heavy-tailed distribution.

¹cf. Chapter 8.3.1 in (Bengio et al., 2017)

²See e.g. (Thorpe, 2018)

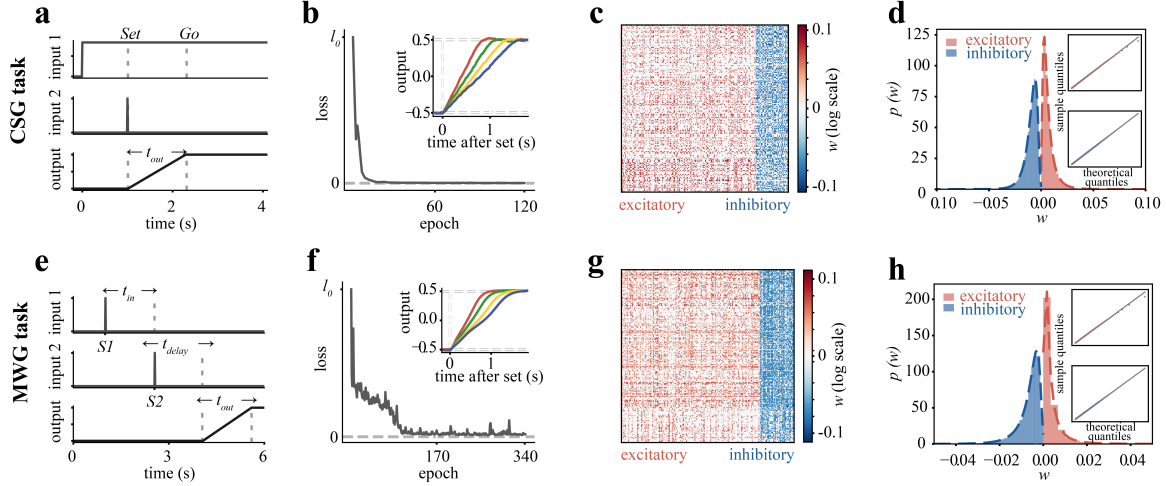


Figure 1. Training RNNs with heavy-tailed connectivity on CSG task and MWG task. (a, e) Structures of the tasks. (b, f) Loss function with respect to the epochs. Here, l_0 means the initial loss value. Inset: Output of heavy-tailed RNNs on four training samples. (c, g) Heat map of neural connectivity matrices. (d, h) Weight distributions of excitatory and inhibitory units and the corresponding Quantile-Quantile plots for the insets.

The following convergence result is established in a simplified optimization setting and is intended to illustrate the stability of the asymptotic DScO-SGD, rather than to provide guarantees for non-convex RNN training. The corresponding proof is provided in Appendix A.3.

Theorem 2.1. Consider a sequence $\{J_n\}_{n \in \mathbb{N}}$ generated by Algorithm 1 with learning rates $0 < \eta_i < 1$, $i = 1, 2$. Assuming the loss function satisfies some assumptions³, it follows that the sequence J_n converges to a limit matrix J^* that satisfies conditions (C1)-(C3).

To validate Algorithm 1, we train heavy-tailed RNNs (cf. Fig. 1 and 9) on a wide range of cognitive tasks (see Table 2 in Appendix A.3), including Cue-Set-Go (CSG) task (Wang et al., 2018; Zhou et al., 2020), Measure-Wait-Go (MWG) task (Beiran et al., 2023), Flip-flop task (Sussillo & Barak, 2013), Mante task (Mante et al., 2013), Romo task (Romo et al., 1999) and perceptual decision making task (Newsome et al., 1989). The convergence of the algorithm is evidenced by the loss trajectories shown in (Fig. 1b, 1f and 9II). Visual inspection of the connectivity heat map (Fig. 1c, 1g and 9III) at the end of training verifies that the constraints of no self-coupling and Dale’s principle are satisfied. Moreover, the quantile-quantile analysis (Fig. 1d, 1h and 9IV) confirms that the learned weights maintain a Log-Normal profile.

3. Heavy-tailed connectivity improves generalization

To assess the influence of heavy-tailed connectivity on generalization, we compared the performance of RNNs trained

Algorithm 1 Asymptotic DScO-SGD

Input: initialization of the weight matrix (J^E, J^I) , task trials $\{u^{(q)}, \hat{z}^{(q)}\}_{q=1, \dots, N_{tr}}$

Parameter: network size N , learning rate (η_1, η_2) , max iteration, task parameters

Output: readout z of the network

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1: Sign-condition = False
2: while  $z^{(q)} \neq \hat{z}^{(q)}$  for some trial  $q$  do
3:   compute empirical risk  $\mathcal{L}(J)$ .
4:   step along the gradient:
      $J_n^E \leftarrow J_n^E - \eta_1^E \nabla \mathcal{L}$ ,  $J_n^I \leftarrow J_n^I - \eta_1^I \nabla \mathcal{L}$ 
5:   if not Sign-condition then
6:     if  $\frac{\#\{J_n^E \neq 0\}}{\#\{J_{n-1}^E \neq 0\}}$  and  $\frac{\#\{J_n^I \neq 0\}}{\#\{J_{n-1}^I \neq 0\}}$  belong to  $(1 - \iota, 1)$  given
       a sufficiently small  $\iota > 0$  then
7:       Sign-condition  $\leftarrow$  True
8:     end if
9:   end if
10:  if Sign-condition then
11:     $\theta_n^E \leftarrow \text{MLE}\{J_n^E\}$ ,  $\theta_n^I \leftarrow \text{MLE}\{J_n^I\}$ 
12:     $\{\hat{J}_n^E\} \leftarrow \rho^m(\theta_n^E)$ ,  $\{\hat{J}_n^I\} \leftarrow \rho^l(\theta_n^I)$ 
13:    sort the weights in ascending order:
      $\{\hat{J}_{(n)}^E : \hat{J}_{(1)}^E \leq \dots \leq \hat{J}_{(fN)}^E\} \leftarrow \{\hat{J}_n^E\}$ 
      $\{\hat{J}_{(n)}^I : \hat{J}_{(1)}^I \leq \dots \leq \hat{J}_{((1-f)N)}^I\} \leftarrow \{\hat{J}_n^I\}$ 
      $\{J_{(n)}^E : J_{(1)}^E \leq \dots \leq J_{(fN)}^E\} \leftarrow \{J_n^E\}$ 
      $\{J_{(n)}^I : J_{(1)}^I \leq \dots \leq J_{((1-f)N)}^I\} \leftarrow \{J_n^I\}$ 
14:    step along the optimal transport map:
      $J_{(n)}^E \leftarrow J_{(n)}^E - \eta_2^E (J_{(n)}^E - \hat{J}_{(n)}^E)$ 
      $J_{(n)}^I \leftarrow J_{(n)}^I - \eta_2^I (J_{(n)}^I - \hat{J}_{(n)}^I)$ 
15:  end if
16:  if exceeds max iteration then
17:    break
18:  end if
19: end while
    
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³See Appendix A.3

with heavy-tailed connectivity on two flexible timing and working memory tasks, CSG and MWG tasks (Fig. 1a and 1e), with that of four baseline models (Table 1). Briefly, in the CSG task, the subject needs to produce different lengths of time intervals according to input amplitudes, which requires a flexible timing function; while in the MWG task, the subject needs to first perceive a given time interval, retain it in working memory and then reproduce it accordingly in motor response, which thus requires a flexible computation for both timing and working memory. Thus, these two tasks provide a minimal set of cognitive functions—specifically, dynamic timing and static working memory—serving as a test bed to examine the impact of heavy-tailed connectivity on generalization.

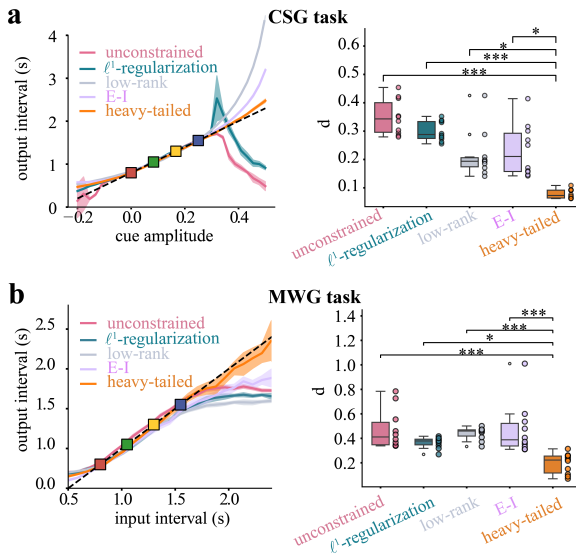


Figure 2. Generalization performance of trained RNNs with different connectivity patterns. Left: characterizing generalization in terms of input-output functions. The dashed line denotes the linear target for the input interval, and proximity to the dashed line corresponds to superior generalization capability. Right: the distances d to the dashed lines computed to quantify generalization, with mean and standard deviation (SD) over 10 trained RNNs, and results of Dunn’s test to show statistically significant difference, in which * means p -value < 0.05 , ** means p -value < 0.01 , and *** means p -value < 0.001 .

For the CSG tasks, we observed that while RNNs of all five models successfully interpolate within the range of trained-cue-amplitudes (Fig. 2a), the RNN with Log-Normal connectivity exhibits significantly superior extrapolation performance. Consistent trends were observed for the MWG task (Fig. 2b). Note that (Beiran et al., 2023) has shown, in explicitly trained low-rank RNN, contextual inputs parametrically controlled the input-output transform, thereby enabling generalization across context values for the MWG task. However, heavy-tailed RNNs achieve better generalization on the MWG task without the need for additional

tonic inputs (Fig. 2b). By pruning the weights located in the heavy tail, we further demonstrated that these values are critical for generalization (Fig. 10). Additionally, experiments with Pareto-distributed weights (Fig. 11) yield similar improvements, confirming that the benefit derives from heavy-tailedness per se rather than the specific Log-Normal form. Furthermore, Fig. 12 establishes a strong correlation between generalization performance and the excess kurtosis (the degree of tailedness) of the weight distributions.

Taken together, these findings show that, in the flexible timing and working memory tasks considered, RNNs endowed with heavy-tailed connectivity are associated with improved generalization performance. We will provide a detailed explanation of this empirical result in Section 4 and 5.

To gain a more in-depth quantitative characterization of the link between heavy-tailed connectivity and generalization performance, we use Log-Normal distributions (Fig. 1d and 1h) as priors, and introduce a modified optimization algorithm, DScO-SGD with prior (see Algorithm 2 in Appendix A.3) to train heavy-tailed RNNs on the CSG and MWG tasks, resulting in final weights that adhere to the target Log-Normal statistics.

An evaluation of their generalization performance (Fig. 3a and 3e) reveals that these newly trained heavy-tailed RNNs exhibit generalization capabilities comparable to those illustrated in Fig. 2. We further observe that the generalization error and Kullback–Leibler (KL) divergence evolve in a correlated manner across epochs (Fig. 3b, 3c and 3f, 3g).

To provide a statistical interpretation of these empirical observations, we analyze generalization bounds within the PAC-Bayes framework⁴. The following theorem provides an interpretative link between generalization performance and the alignment between posterior and prior distributions (proof given in Appendix A.5).

Theorem 3.1. *Let $\beta \in (0, 1)$. Then with probability at least $1 - \beta$ over dataset $S = \{u^{(q)}, \hat{z}^{(q)}\}_{q=1, \dots, N_{tr}}$, we have*

$$\mathbb{E}[\varepsilon] \lesssim \sqrt{\frac{\text{KL}(\rho_S || \pi) + \log \frac{3}{\beta}}{N_{tr}}}$$

where π is a heavy-tailed prior distribution, ρ_S as posterior distribution, and $\text{KL}(\cdot || \cdot)$ denotes the Kullback–Leibler divergence.

Given N_{tr} training trials, Theorem 3.1 suggests a PAC-Bayesian interpretation for the generalization performance: solutions whose posterior distributions are more closely aligned with the prior exhibit lower statistical complexity,

⁴cf. (Alquier, 2024). This is a useful framework in machine learning for deriving generalization bounds, blending elements of Bayesian methods with the Probably Approximately Correct (PAC) learning theory.

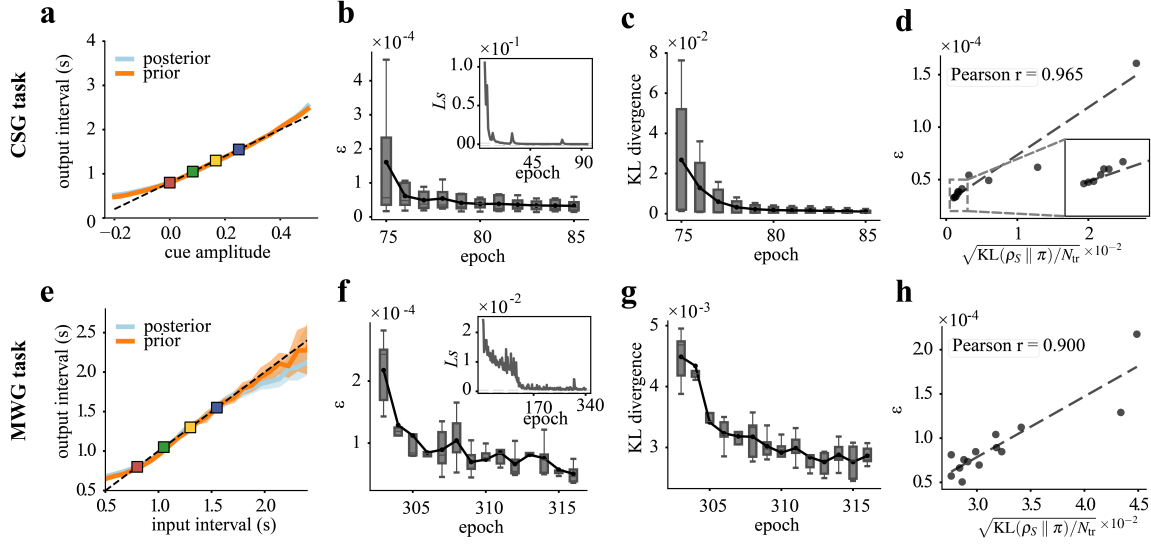


Figure 3. Relationship between generalization error and KL divergence. (a, e) Generalization capabilities of trained RNNs by DScO-SGD with priors. prior: generalization of the heavy-tailed RNN from Fig. 1. posterior: RNN trained by Algorithm 2 with Log-Normal prior (shown in Fig. 1d and 1h). (b, f) Training and generalization errors with respect to the epochs. (c, g) KL divergence between posterior and prior distributions of neural connectivities during training. (d, h) Pearson correlation between the generalization error and the Kullback–Leibler (KL) divergence. The p -values for panels (d) and (h) are 1.486×10^{-6} and 1.1634×10^{-5} , respectively.

which correlates with improved generalization performance. In particular, the empirical results (Fig. 3d and 3h) show the linear correlation between ε and $\sqrt{\frac{\text{KL}(\rho_S \parallel \pi)}{N_{tr}}}$, which are consistent with this theoretical derivation.

4. Heavy-tailed connectivity induces low-dimensional dynamics through effective low-rank structure

Thus far, we have empirically observed that heavy-tailed connectivity facilitates the generalization of RNNs. What would be the underlying mechanisms driving this empirical phenomenon? Previous studies (Mastrogiuseppe & Ostojic, 2018; Beiran et al., 2023) demonstrated that low rank connectivity induces low-dimensional dynamics, which might lead to good generalizations. Following this line of studies, in this section, we investigate how heavy-tailed connectivity is associated with the emergence of an effective low-rank structure (see Fig. 4a and 4e, Table 3), which further gives rise to low-dimensional dynamics of neural activity (cf. Theorem 4.4 and Fig. 4d and 4h).

Before giving our main result, we need some preparations.

Let A be an $N \times N$ random matrix. Denote the singular values of A by

$$\gamma_1 \geq \gamma_2 \geq \dots \gamma_N \geq 0.$$

Define the relative spectral gap as

$$g_k(A) = \frac{\gamma_k - \gamma_{k+1}}{\gamma_k}, \quad k = 1, 2, \dots, N-1.$$

In addition, we define the following quantities to measure the latent structure of matrix A .

- Mathematical rank $r_m(A) = \max\{k : \gamma_k > 0\}$.
- Shannon effective rank (cf. (Roy & Vetterli, 2007))

$$r_{\text{eff}}(A) = \exp \left\{ - \sum_{k=1}^N q_k \ln q_k \right\}$$

$$\text{with } q_k = \frac{\gamma_k^2}{\|J\|_F^2} = \frac{\gamma_k^2}{\sum_{k=1}^N \gamma_k^2}.$$

It is easy to see that $r_m \leq N$. Due to the convex property of function $-\ln(\cdot)$, it follows from Jensen’s inequality that

$$- \sum_{k=1}^N q_k \ln q_k = \sum_{k=1}^{r_m} q_k (-\ln q_k) \leq \ln r_m,$$

where equality holds if and only if $q_k = \frac{1}{r_m}$ for all $k = 1, \dots, r_m$. Finally, we obtain $r_{\text{eff}} \leq r_m \leq N$.

For the Gaussian case, it implies from the following lemma that the gaps are uniformly small and exhibit no pronounced jumps.

Lemma 4.1. Suppose the entries A_{ij} of A are independently identically distributed (i.i.d.) drawn from a Gaussian distribution $\mathcal{N}(0, 1)$. Then the effective rank converges to a constant strictly less than 1, i.e.

$$\lim_{N \rightarrow \infty} \frac{r_{\text{eff}}(A)}{N} = c_0 < 1, \quad \text{almost surely.}$$

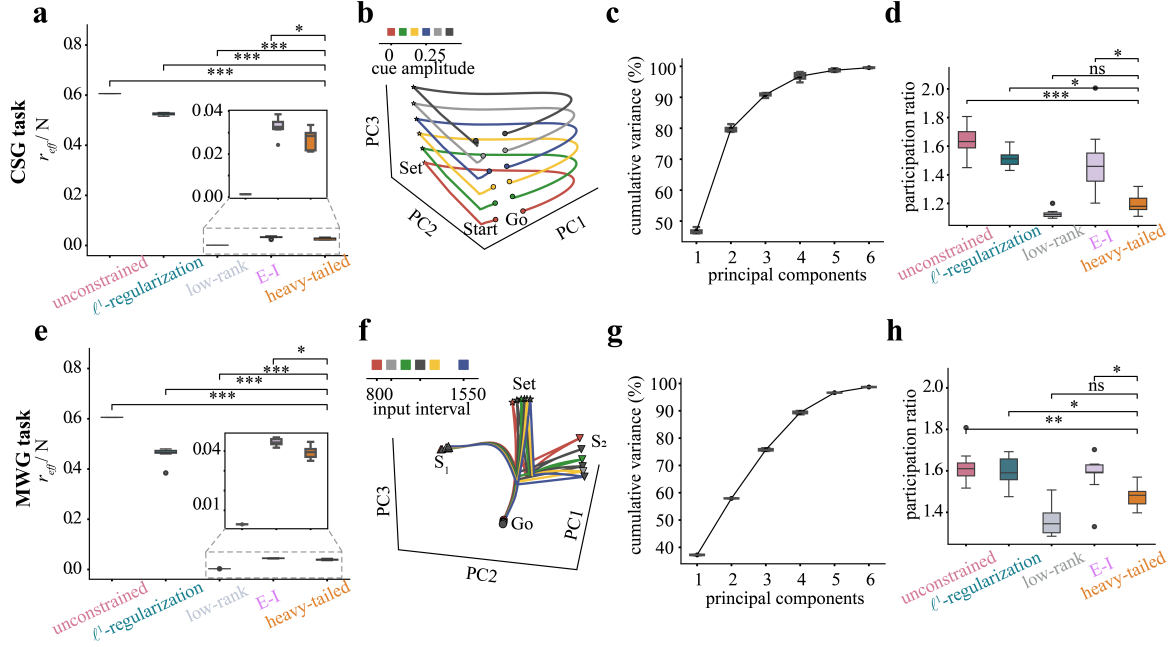


Figure 4. Investigating Shannon effective rank of the heavy-tailed connectivity and dimensionality of the neural dynamics. (a, e) Effective ranks of different neural connections with mean and SD over 10 trained RNNs. (b, f) Neural activities generated by the heavy-tailed RNNs (Fig. 1) in the subspace spanned by the three leading principal components in the standard principal component analysis (PCA). (c, g) Cumulative percentage of explained variance in the PCA. (d, h) Participation ratio of activity during ‘Set’ and ‘Go’, with mean and SD over 10 trained RNNs. Wilcoxon rank-sum test: * means p -value < 0.05 , ** means p -value < 0.01 , *** means p -value < 0.001 , and ns means no significance.

where c_0 is governed by the entropy of the Marchenko–Pastur distribution. Moreover, in the bulk of the spectrum, the relative gaps $g_k(A)$ scales as $g_k \sim O(\frac{1}{N})$, and the fluctuation between the maximum and minimum gaps vanishes, that is, $\max_k g_k - \min_k g_k = o(1)$, w.h.p.

The following lemma implies that the existence of large gaps w.h.p for the heavy-tailed case.

Lemma 4.2. Assume that the entries A_{ij} of A are i.i.d. drawn from a heavy-tailed distribution satisfying the regular variation condition with tail index $\alpha > 0$, that is,

$$\mathbb{P}(|A_{ij}| > x) = L(x)x^{-\alpha}, \quad \text{as } x \rightarrow \infty,$$

where $L(x)$ is a slowly varying function⁵. For a small index k (e.g., $k = 1$), the relative spectral gap g_k does not vanish as $N \rightarrow \infty$ but it remains bounded away from zero with high probability. Additionally, if $A_{ij} \sim \text{LogNormal}(0, \sigma^2)$ with some fixed $\sigma > 0$, then for any $\delta \in (0, 1)$ there exist a positive constant $\hat{\alpha}$, depending on σ , such that, with probability at least $1 - \delta$, one has a small index $k^* = O(1)$, independent of N , satisfies $g_{k^*} \geq 1 - e^{-\hat{\alpha}\sigma^2}$.

Moreover, if $|A_{ij}|$ follows a heavy-tailed distribution with tail index $\alpha \in (0, 2)$ (Infinite variance regime), $\frac{r_{\text{eff}}(A)}{N}$ converges to 0 in probability as N goes to infinity. For the

lognormal case, there exists $0 < c < 2\hat{\alpha}$ such that

$$r_{\text{eff}}(A) \leq 2k^* + O(N \cdot e^{-c\sigma^2}).$$

with high probability as $N \rightarrow \infty$.

The proof of Lemma 4.1 and 4.2 can be found in Appendix A.6 and Appendix A.7, respectively. Let us introduce two $N \times N$ matrices defined by

$$\tilde{J}^E = [J^E \quad \mathbf{0}_{N \times (1-f)N}], \quad \tilde{J}^I = [\mathbf{0}_{N \times fN} \quad J^I].$$

Note that

$$r_{\text{eff}}(J^E) = r_{\text{eff}}(\tilde{J}^E), \quad r_{\text{eff}}(J^I) = r_{\text{eff}}(\tilde{J}^I)$$

and $r_{\text{eff}}(J) \leq r_{\text{eff}}(J^E) + r_{\text{eff}}(J^I)$. Since the trained weight matrix is not strictly i.i.d. and correlations further reduce effective rank ((Roy & Vetterli, 2007)), then applying Lemma 4.2 to the heavy-tailed connection J yields one of our main theorems as follows.

Theorem 4.3. For a connectivity matrix J characterized by heavy-tailed weights (e.g., infinite variance regimes or Log-Normal distributions), we have $1 \leq r_{\text{eff}}(J) \ll N$ with high probability as $N \rightarrow \infty$.

This theorem explains why trained heavy-tailed connectivity (despite being non-i.i.d.) tends to exhibit a low effective rank

⁵cf. Definition 2.3 (Nair et al., 2022)

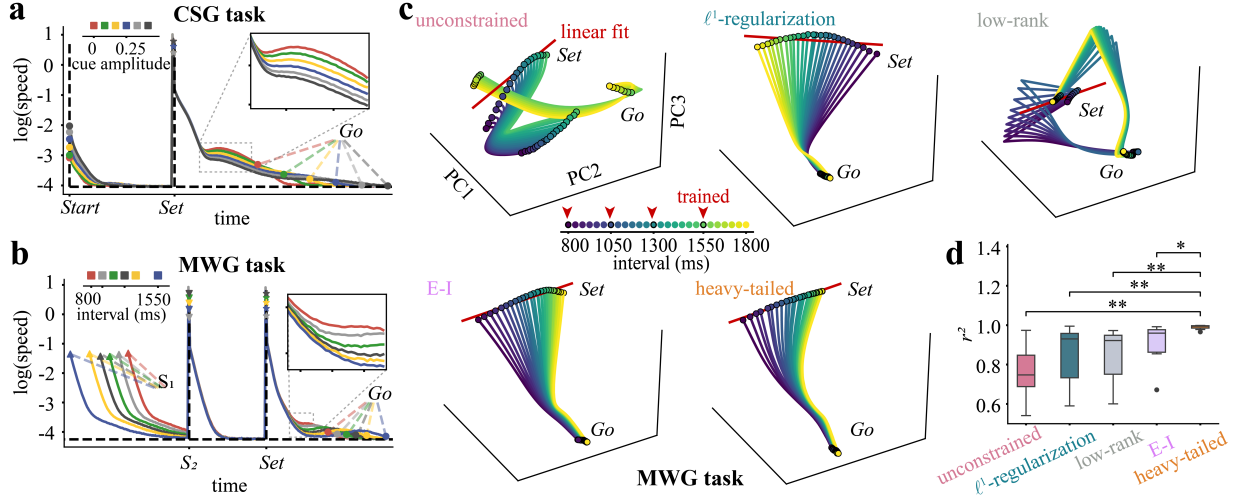


Figure 5. Dynamics speeds and representational geometry. (a, b) The speed during the set and go signals is scaled by the cue amplitude (CSG task) or input interval (MWG task), even beyond the training range. (c) The dynamics in the embedding subspace of trained RNNs with different connectivities during the ‘Set’ and ‘Go’ signals exhibit distinct manifold geometry. Red lines denote the linear fits of the states at the ‘Set’ phase. (d) Quantification of the linear representational geometry at the ‘Set’ phase. r^2 is the coefficient of determination of the linear fits, i.e. the square of the Pearson coefficient, with mean and SD over 10 trained RNNs. Wilcoxon rank-sum test: * means p -value < 0.05 , ** means p -value < 0.01 , *** means p -value < 0.001

in practice (Fig. 4a and 4e, Table 3). Moreover, the trained heavy-tailed connectivity matrices (with or without element-wise shuffle) possess a significantly lower effective rank compared to that of their Gaussian counterparts (Fig. 13).

We then seek to demonstrate whether the low effective rank induced by the heavy-tailed connectivity gives rise to low-dimensional dynamics. To quantify the dimensionality of neural activity, besides the variance explained by the first principal component, we also looked at a complementary approach, the participation ratio. First, we compute the empirical covariance matrix $C_{ij} = \mathbb{E}[\varphi(x_i)\varphi(x_j)]$ of the neural signal $\varphi(x_j)$. Let λ_k denote the eigenvalues of the covariance $\{C_{ij}\}$. The participation ratio is defined as

$$r_{\text{pr}}(\{C_{ij}\}) = \frac{(\sum_{k=1}^N \lambda_k)^2}{\sum_{k=1}^N \lambda_k^2}.$$

Then we have the following theorem.

Theorem 4.4. *If the connection J satisfies $r_{\text{eff}}(J) = o(N)$, then $r_{\text{pr}} \ll N$, which implies that the dynamics governed by (1) is low-dimensional.*

The proof of this theorem can be found in Appendix A.8. To visualize the low dimensionality of the dynamics, we applied principal component analysis to the firing rates $\varphi(x_j)$, $j = 1, \dots, N$, of the recurrent units in given tasks. We then projected the neural activity onto the low-dimensional subspace spanned by the leading principal components (Fig. 4).

For different amplitudes of the cue in the CSG task, we found that the neural trajectories evolved along parallel

curves (Fig. 4b), whereas for different input intervals in the MWG task, the trajectories followed more complex paths due to the more complex task structure (Fig. 4f and Fig. 14). However, the dynamics during the same task period (e.g. between S1 and S2) tends to traverse parallelly with each other in a low-dimensional regime as well. Strikingly, the geometric shapes of these trajectories remained largely invariant across different task conditions (e.g., different cue amplitudes or input intervals).

The dimensionalities of neural dynamics following the set signal for RNNs with distinct architectures (see Table 1) on CSG and MWG tasks are presented in (Fig. 4d and 4h), corroborating our theoretical result in Theorem 4.4. Note that RNNs with heavy-tailed connectivity achieve similar effective low rank and low dimensionality as the RNNs explicitly trained with three ranks constraint, and significantly lower rank and dimensionality than the other model (Fig. 4).

5. Heavy-tailed connectivity induces linear manifold representations of task variables

The results presented above demonstrate that RNNs with heavy-tailed connections acquire low-dimensional representations for both the CSG and MWG tasks (Fig. 4). However, low dimensionality per se is insufficient to guarantee generalization, as shown by the fact that explicitly trained low rank RNNs exhibit similar low dimensionality as heavy-tailed RNNs, but performed worse in the generalization (Fig. 2). This raises a critical question: whether there is an extra intrinsic mechanism induced by the heavy-

tailed connectivity, functionally analogous to a tonic input in (Beiran et al., 2023), to regulate the overall input-output mapping.

We addressed this question by first investigating the speed of the neural dynamics. In the CSG task, we found that the speed of the dynamics along each curve during the ‘Set’ and ‘Go’ phase is parametrically modulated by the cue amplitude (Fig. 5a, inset). In the MWG task, although the geometry of the paths is complex, we observed that the speed adapts to distinct input intervals (Fig. 5b, inset). This provides compelling evidence that heavy-tailed RNNs learn intrinsic modulations of dynamics speed, obviating the need for external control signals required by explicitly-trained low-rank RNNs (Beiran et al., 2023).

Moreover, we observed that during the ‘Set’ and ‘Go’ phases, RNNs with different architectures, trained on the MWG task, evolve along distinct manifolds (Fig. 5c). Notably, heavy-tailed connections effectively govern the gain of the recurrent dynamics, allowing the input interval to scale the flow speed along the low-dimensional manifold. Furthermore, dynamics of the heavy-tailed RNNs form a more linear manifold representation at the ‘Set’ phase (Fig. 5d). Intuitively, these linear manifold representations provide a relatively uniform coding of time intervals into the memory, which in turn facilitates the generalization for unseen intervals in a commonly used linear readout strategy. While a complete theoretical derivation of the intrinsic modulation of dynamics speed and linear manifold representation by heavy-tailed connectivity remains an open challenge due to the nonlinearity of the system, we hypothesize that the heavy-tailed weights might act as a ‘gating mechanism’, which allows the dynamics flow in a specific ‘optimal tunnel’. It is worth noting that gated recurrent units (GRUs, cf. (Haykin, 2009)) control dynamics speed by explicitly modulating the interpolation between past and present hidden states, which is similar to that observed in heavy-tailed RNNs (Fig. 5a and 5b). This suggests that GRUs, or simplified variants possessing only an update gate (Collins et al., 2017), might naturally develop more linear manifolds than vanilla RNNs owing to their built-in speed modulation. The defining advantage of our heavy-tailed RNN, however, is that it realizes this sophisticated speed modulation entirely without explicit gating architectures, relying solely on the emergent properties of its weight statistics.

6. Conclusion and discussion

By introducing a novel optimization framework, DScot-SGD, we successfully trained RNNs that satisfy key biological constraints, including Dale’s principle and heavy-tailed weight statistics. Our findings reveal that heavy-tailed connectivity naturally induces an effective low rank structure,

compresses the neural trajectories into low-dimensional manifolds, favors a linear manifold representation of task variables, and in turn enhances generalizations in flexible timing and working memory tasks (see Fig. 6).

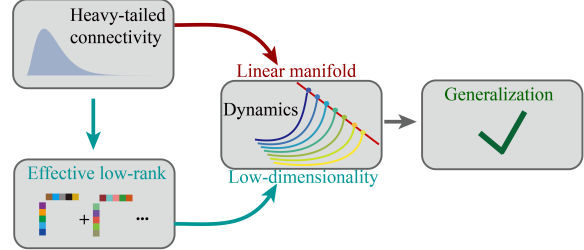


Figure 6. A conceptual schematic.

A key theoretical insight of this study is the link between heavy-tailed statistics and the emergence of low-rank structure. Intuitively, while the spectral distribution of a Gaussian random matrix tends to be compact with vanishing spectral gaps as network size increases—effectively behaving like a ‘zero matrix’ or structureless noise under high-threshold filtering, heavy-tailed matrices exhibit a fundamentally different behavior. The presence of outlier weights in the heavy tail naturally separates the eigenspectrum, preserving a ‘skeleton’ of strong connections even when small weights are pruned. This mathematically corresponds to a sparse, effective low-rank approximation that remains robust even in large-scale networks. Consequently, this structural rigidity forces the rich dynamics of the RNN to collapse into a low-dimensional subspace, providing a stable geometric basis for computation.

This emergence of low-dimensional dynamics aligns our work with recent findings by (Beiran et al., 2023), who demonstrated that low-rank RNNs can support flexible cognitive computations. However, a crucial distinction lies in the origin of this structure. While (Beiran et al., 2023) explicitly constructed low-rank connectivity via low-dimensional parameterization, our work shows that such structure can emerge spontaneously from biologically plausible heavy-tailed statistics without enforcing a low-rank bottleneck during training. Furthermore, regarding the mechanism of generalization, previous studies demonstrated that external mechanisms are needed to parametrically control the network dynamics: (Beiran et al., 2023) and (Zhou et al., 2022) rely on external tonic inputs, while (Zhou & Buonomano, 2024) relies on an external neural modulation of short-term plasticity. In contrast, our heavy-tailed RNNs exhibit an intrinsic modulation mechanism. We posit that the strong connections in the heavy tail act as gating factors, spontaneously forming a linear manifold where task variables (such as time intervals) are linearly encoded. Since

linear readouts from linear manifolds are known to favor extrapolation, this intrinsic geometry likely underpins the superior generalization we observed. Importantly, ‘linearity’ in our context defines a representational structure optimized for smooth parametric extrapolation. This contrasts with recent approaches, such as in (Qian & Pehlevan, 2026), which explicitly penalize linear predictivity to force the discovery of alternative, dynamic solutions. Ultimately, unraveling why heavy-tailed statistics specifically favor the formation of linear manifolds remains a compelling mathematical question for future research.

Our results also bridge the gap between statistical mechanics theories of neural networks and functional generalization. Consistent with (Xie et al., 2025), who reported that heavy-tailed weights reduce the Lyapunov dimension of chaotic attractors, we confirm that heavy tails lead to lower effective dimensionality. Moreover, our empirical findings resonate with (Simsekli et al., 2020), who theoretically linked the tail index of weight distributions to generalization error in feedforward networks. We extend these perspectives by demonstrating how this dimensionality reduction translates into functional advantages in dynamical timing and working memory tasks. Unlike previous works that focus primarily on the statistical properties of the weights, we identify the specific representational geometry and topology (i.e., linear manifold representations of task variable and overall low-dimensional dynamics) as the dynamical bridge between heavy-tailed structure and generalization performance.

Finally, we acknowledge that our DScO-SGD algorithm, which relies on global gradients and optimal transport, is not biologically plausible. The algorithm serves as a means to an end: identifying the functional ‘what’ (heavy-tailed connectivity) rather than the developmental ‘how’. A promising direction for future research is to investigate how such heavy-tailed distributions can arise from biologically realistic local learning rules. For instance, exploring three-factor Hebbian learning rules (Kuśmierz et al., 2017) reinforced by neuromodulation could offer a mechanistic explanation for how the brain self-organizes into these generalization-promoting structures.

Acknowledgements

Z. Jiao’s research is supported by the National Natural Science Foundation of China (12272297). S. Zhou was supported by the Natural Science Foundation of Shanghai (24ZR1414800), and the National Natural Science Foundation of China (32571217).

Impact Statement

This work offers a theoretical and algorithmic exploration at the intersection of machine learning and computational

neuroscience. Its primary impact is foundational, advancing our understanding of how bio-plausible constraints shape connection structure, representation geometry and network generalization. As a piece of fundamental research focused on the topological and geometrical properties of neural dynamics, its sociological implications are long-term and indirect. We do not foresee any direct ethical concerns, immediate negative societal impacts, or specific risks of misuse requiring exceptional highlighting here.

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A. Appendix

A.1. Notations

- $\#\{\cdot\}$ means the number of the elements in the set.
- $f(n) = O(g(n))$. There exist constants $C > 0$ and n_0 such that

$$|f(n)| \leq Cg(n)$$

for any $n \geq n_0$.

- $f(n) = o(g(n))$. For any $\epsilon > 0$, there exists a constant n_0 such that

$$|f(n)| \leq \epsilon g(n)$$

when $n \geq n_0$.

- An event E holds with high probability (w.h.p) if it holds with probability $1 - O(\frac{1}{n^c})$ for some constant $c > 0$ independent of n that is, one has

$$\mathbb{P}(E) \geq 1 - \frac{C}{n^c}$$

for some C independent of n .

A.2. RNNs with different connectivities

Table 1. RNNs with different neural connections and selective references. For low-rank RNN, we use the same model as in (Beiran et al., 2023), where the random part of the connectivity matrix is equal to zero.

RNN	Connectivity	C1	C2	C3	References
Unconstrained	/	✗	✗	✗	(Haykin, 2009)
l^1 -regularization	Sparse	✗	✗	✗	(Haykin, 2009)
Low-rank	Spiked-covariance	✗	✗	✗	(Mastrogiuseppe & Ostojic, 2018)
E-I	Dale’s principle	✓	✓	✗	(Song et al., 2016)
Heavy-tailed	Log-Normal	✓	✓	✓	This work

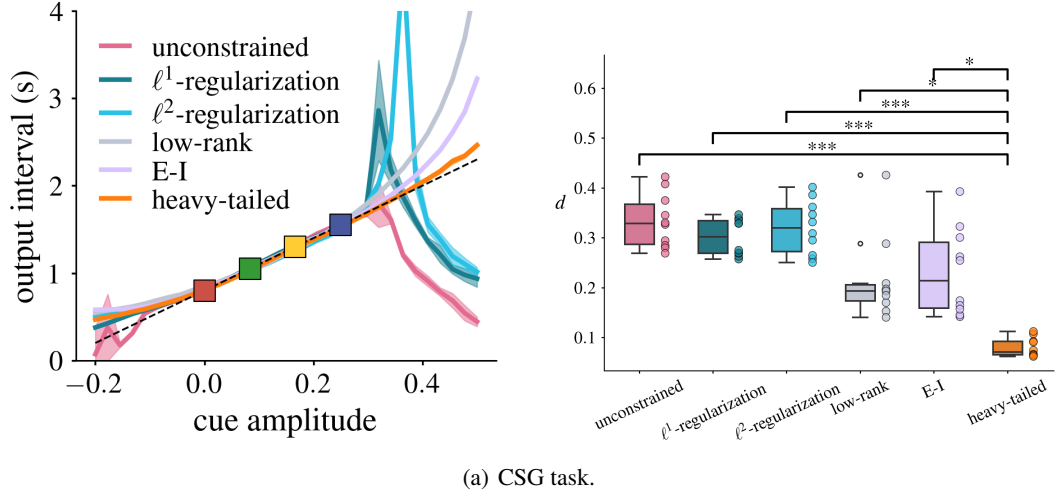
These baselines were selected to investigate specific scientific mechanisms, rather than merely for performance benchmarking:

- Unconstrained: Serves as the standard, unregularized baseline.
- l^1 -regularization: Tests whether weight sparsity alone is sufficient to achieve the observed benefits, without relying on heavy-tailed distributions.
- Low-rank: Represents the most relevant computational neuroscience baseline (Beiran et al., 2023), allowing us to directly compare explicit versus emergent low-rank structures.
- E-I: Serves as an ablation to evaluate the isolated effect of Dale’s principle in the absence of heavy-tailed connectivity.

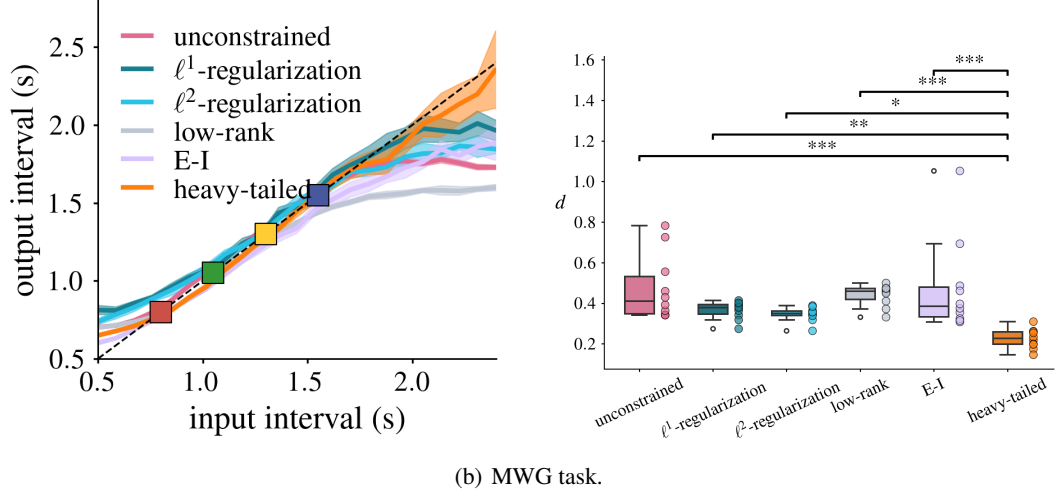
As suggested by one of the reviewers, Fig. 7 compares the generalization performance of the ℓ_2 -regularization baseline with the other baselines presented in Table 1. These results are entirely consistent with our main conclusions.

To provide a more comprehensive ablation study, we have conducted the following additional experiments:

- Heavy-tailed without Dale’s principle (Fig. 8, Left): to isolate the effect of the distributional constraint from the sign constraint.
- Heavy-tailed with self-connection (Fig. 8, Right): to test the role of self-coupling $J_{ii} = 0$.

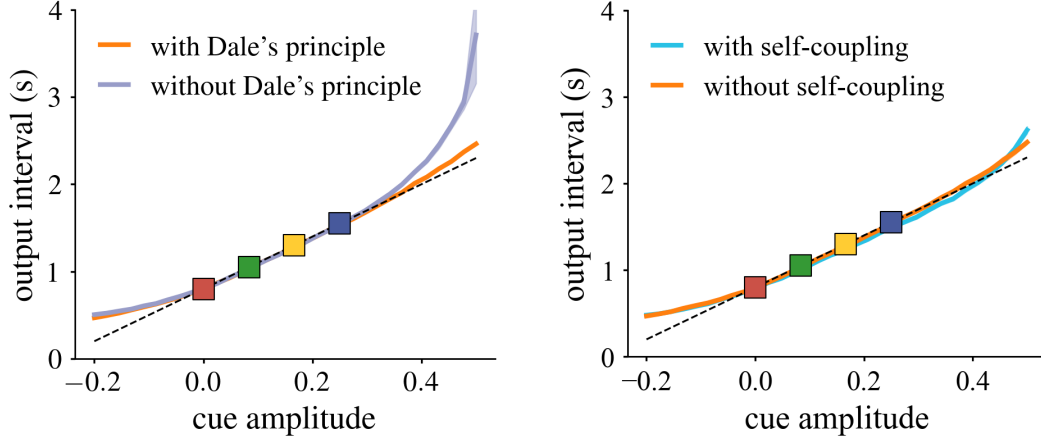


(a) CSG task.

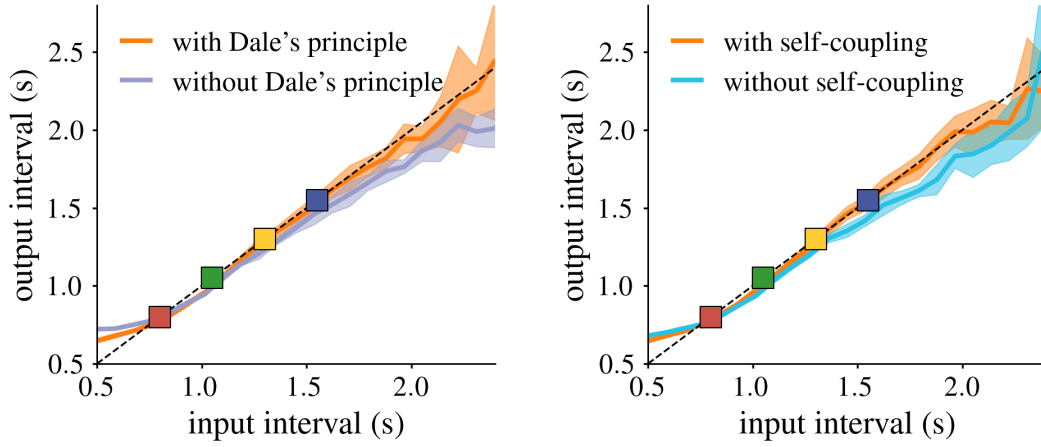


(b) MWG task.

Figure 7. Generalization performance of trained RNNs with different connectivity patterns.



(a) CSG task.



(b) MWG task.

Figure 8. Complete ablation of heavy-tailed RNNs. Left: to isolate the effect of the distributional constraint from the sign constraint. Right: to test the role of self-connection.

A.3. Convergence of Algorithms

We propose two algorithms to solve the minimization problem

$$\min_{J \in \mathbb{R}^{N \times N}} \mathcal{L}_S(J).$$

Asymptotic DScO-SGD (Algorithm 1) is to find a heavy-tailed connection matrix $J \in \operatorname{argmin} \mathcal{L}_S(\mathcal{J})$, while DScO-SGD with prior distributions (see Algorithm 2) is to find a connection matrix $J \in \operatorname{argmin} \mathcal{L}_S(\mathcal{J})$, which is drawn from pre-specified heavy-tailed distributions. It is worth noting that the proof of convergence for Algorithm 2 is analogous to that of Algorithm 1. Subsequently, we just give the proof of asymptotic DScO-SGD.

Algorithm 2 DScO-SGD with priors

Input: initialization of the weight matrix (J^E, J^I) , task trials $\{u^{(q)}, \hat{z}^{(q)}\}_{q=1, \dots, N_{\text{tr}}}$

Parameter: network size N , learning rate (η_1, η_2) , max iteration, task parameters, two prior distributions $\hat{\rho}(\hat{\theta}^E)$ and $\hat{\rho}(\hat{\theta}^I)$

Output: readout z of the network

```

1: Sign-condition = False
2:  $\{\hat{J}_i^E\} \leftarrow \hat{\rho}(\hat{\theta}^E), \{\hat{J}_i^I\} \leftarrow \hat{\rho}(\hat{\theta}^I)$ 
3: arrange the weights in the ascending order:
    $\{\hat{J}_{(i)}^E : \hat{J}_{(1)}^E \leq \dots \leq \hat{J}_{(fN)}^E\} \leftarrow \{\hat{J}_i^E\}$ 
    $\{\hat{J}_{(i)}^I : \hat{J}_{(1)}^I \leq \dots \leq \hat{J}_{((1-f)N)}^I\} \leftarrow \{\hat{J}_i^I\}$ 
4: while  $z^{(q)} \neq \hat{z}^{(q)}$  for some trial  $q$  do
5:   compute empirical risk  $\mathcal{L}(J)$ .
6:   step along the gradient:
      $J_i^E \leftarrow J_i^E - \eta_1^E \nabla \mathcal{L}, \quad J_i^I \leftarrow J_i^I - \eta_1^I \nabla \mathcal{L}$ 
7:   if not Sign-condition then
8:     if  $\frac{\#\{J_n^E \neq 0\}}{\#\{J_{n-1}^E \neq 0\}}$  and  $\frac{\#\{J_n^I \neq 0\}}{\#\{J_{n-1}^I \neq 0\}}$  belong to  $(1 - \iota, 1)$  given a sufficiently small  $\iota > 0$  then
9:       Sign-condition  $\leftarrow$  True
10:    end if
11:  end if
12:  if Sign-condition then
13:    arrange the weights in the ascending order:
      $\{J_{(i)}^E : J_{(1)}^E \leq \dots \leq J_{(fN)}^E\} \leftarrow \{J_i^E\}$ 
      $\{J_{(i)}^I : J_{(1)}^I \leq \dots \leq J_{((1-f)N)}^I\} \leftarrow \{J_i^I\}$ 
14:    step along the optimal transport map:
      $J_{(i)}^E \leftarrow J_{(i)}^E - \eta_2^E (J_{(i)}^E - \hat{J}_{(i)}^E)$ 
      $J_{(i)}^I \leftarrow J_{(i)}^I - \eta_2^I (J_{(i)}^I - \hat{J}_{(i)}^I)$ 
15:  end if
16:  if exceeds max iteration then
17:    break
18:  end if
19: end while

```

Assume the loss function $\mathcal{L}$ satisfies the following conditions:

- ν -strongly convex, that is,

$$\nu \frac{t(1-t)}{2} \|J_1 - J_2\|^2 + \mathcal{L}(tJ_1 + (1-t)J_2) \leq t\mathcal{L}(J_1) + (1-t)\mathcal{L}(J_2), \quad \text{for all } J_1, J_2 \in \mathbb{R}^{N \times N}, t \in [0, 1].$$

- L -smooth for some $0 < \nu \leq L$, that is,

$$\|\nabla \mathcal{L}(J_1) - \nabla \mathcal{L}(J_2)\| \leq L \|J_1 - J_2\|, \quad \text{for all } J_1, J_2 \in \mathbb{R}^{N \times N}.$$

We now present the proof of Theorem 2.1.

Proof. The convergence of the asymptotic DSCO-SGD algorithm is governed by the iterations following the satisfaction of the sign condition, which is illustrated by

$$J_n^m \xrightarrow{\text{SGD}} J_{n+1}^m \xrightarrow{\text{Optimal transport}} J_{(n+2)}^m \xrightarrow{\text{Arranging}} J_{n+2}^m,$$

which consists of the SGD update

$$J_{n+1}^m = J_n^m - \eta_1^m \nabla \mathcal{L}_S(J_n^m) \quad (2)$$

and the optimal transport map

$$J_{(n+2)}^m = J_{(n+1)}^m - \eta_2^m \left(J_{(n+1)}^m - \hat{J}_{(n+1)}^m \right). \quad (3)$$

Here, $\hat{J}_{n+1}^m$ is the sample generated by a heavy-tailed distribution whose parameters are derived by the maximum likelihood estimation on J_{n+1}^m , and $J_{(n+1)}^m$ is in ascending order of J_{n+1}^m .

First, we note that (2) is equivalent to

$$J_{(n+1)}^m = J_n^m - \eta_1^m \nabla \mathcal{L}_S(J_n^m).$$

Let J^* be some minimizer and $\hat{J}^*$ in ascending order of J^* . Then we have

$$J_{(n+1)}^m - J^* = J_n^m - J^* - \eta_1^m \nabla \mathcal{L}_S(J_n^m). \quad (4)$$

Since $\mathcal{L}$ is ν -strongly convex and L -smooth, for some $0 < \nu \leq L$, by (4) we have

$$\begin{aligned} & \left\| J_{(n+1)}^m - J^* \right\|^2 \\ &= \left\| J_n^m - J^* - \eta_1^m \nabla \mathcal{L}_S(J_n^m) \right\|^2 \\ &= \left\| J_n^m - J^* \right\|^2 - 2\eta_1^m \left(\nabla \mathcal{L}_S(J_n^m), J_n^m - J^* \right) + (\eta_1^m)^2 \left\| \nabla \mathcal{L}_S(J_n^m) \right\|^2 \\ &\leq (1 - \eta_1^m \nu) \left\| J_n^m - J^* \right\|^2 - 2\eta_1^m (\mathcal{L}_S(J_n^m) - \min \mathcal{L}) + (\eta_1^m)^2 \left\| \nabla \mathcal{L}_S(J_n^m) \right\|^2 \\ &\leq (1 - \eta_1^m \nu) \left\| J_n^m - J^* \right\|^2 - 2\eta_1^m (\mathcal{L}_S(J_n^m) - \min \mathcal{L}) + 2(\eta_1^m)^2 L (\mathcal{L}_S(J_n^m) - \min \mathcal{L}) \\ &\leq (1 - \eta_1^m \nu) \left\| J_n^m - J^* \right\|^2 - 2\eta_1^m (1 - \eta_1^m \nu) (\mathcal{L}_S(J_n^m) - \min \mathcal{L}) \\ &\leq (1 - \eta_1^m \nu) \left\| J_n^m - J^* \right\|^2 \end{aligned} \quad (5)$$

for $0 < \eta_1^m \leq \frac{1}{L}$. From (3) it follows

$$\begin{aligned} J_{(n+2)}^m - \hat{J}^* &= J_{(n+1)}^m - \hat{J}^* - \eta_2^m \left[\left(J_{(n+1)}^m - \hat{J}^* \right) - \left(\hat{J}_{(n+1)}^m - \hat{J}^* \right) \right] \\ &= (1 - \eta_2^m) \left(J_{(n+1)}^m - \hat{J}^* \right) + \eta_2^m \left(\hat{J}_{(n+1)}^m - \hat{J}^* \right), \end{aligned}$$

which implies

$$J_{(n+2)}^m - \hat{J}^* = (1 - \eta_2^m)^{n+2} \left(J_{(0)}^m - \hat{J}^* \right) + \sum_{k=1}^{n+1} \eta_2^m (1 - \eta_2^m)^k \left(\hat{J}_{(n-k+1)}^m - \hat{J}^* \right).$$

We further obtain

$$\left\| J_{(n+2)}^m - \hat{J}^* \right\|^2 \leq (1 - \eta_2^m)^{n+2} \left\| J_{(0)}^m - \hat{J}^* \right\|^2 + \sum_{k=1}^{n+1} \eta_2^m (1 - \eta_2^m)^k \left\| \hat{J}_{(n-k+1)}^m - \hat{J}^* \right\|^2. \quad (6)$$

Note that the following equivalent relationship holds true

$$\left\| \hat{J}_{(n-k+1)}^m - \hat{J}^* \right\|^2 \sim \left\| J_{(n-k+1)}^m - J^* \right\|^2.$$

Due to (5), we have

$$\left\| \hat{J}_{(n-k+1)}^m - \hat{J}^* \right\|^2 \leq (1 - \eta_1^m \nu) \left\| \hat{J}_{(n-k)}^m - \hat{J}^* \right\|^2, \quad \text{for } k = 1, 2, \dots, n+1. \quad (7)$$

Combining (6) and (7) yields

$$\left\| J_{(n+2)}^m - \hat{J}^* \right\|^2 \leq (1 - \eta_2^m)^{n+2} \left\| J_{(0)}^m - \hat{J}^* \right\|^2 + \eta_2^m \sum_{k=1}^{n+1} (1 - \eta_2^m)^k (1 - \eta_1^m \nu)^{n-k+1} \left\| \hat{J}_{(0)}^m - \hat{J}^* \right\|^2.$$

Thanks to $0 < \eta_1^m \leq \frac{1}{L}$, $0 < \eta_1^m \leq 1$ and $0 < \nu \leq L$, we further get

$$\begin{aligned} & \left\| J_{(n+2)}^m - \hat{J}^* \right\|^2 \\ & \leq \begin{cases} (1 - \eta_2^m)^{n+2} \left\| J_{(0)}^m - \hat{J}^* \right\|^2 + \frac{\eta_2^m (1 - \eta_2^m) (1 - \eta_1^m \nu) [(1 - \eta_1^m \nu)^n - (1 - \eta_2^m)^n]}{\eta_2^m - \nu \eta_1^m} \left\| \hat{J}_{(0)}^m - \hat{J}^* \right\|^2, & \eta_1^m \nu \neq 1, \nu \eta_1^m \neq \eta_2^m \\ (1 - \eta_2^m)^{n+2} \left\| J_{(0)}^m - \hat{J}^* \right\|^2, & \eta_1^m \nu = 1, \end{cases} \\ & \rightarrow 0, \quad n \rightarrow \infty, \end{aligned}$$

which implies the convergence of the asymptotic DScO-SGD algorithm. $\square$

Although extending our results to broader classes of loss functions, such as those satisfying the Polyak–Łojasiewicz (PL) condition and smoothness assumptions, is theoretically possible, it remains beyond the scope of this work. However, it is worth noting that the loss function used to train our heavy-tailed RNNs locally satisfies the PL condition and exhibits a one-sided Lipschitz property. As shown in Theorem 3.5 (Wojtowysch, 2024), this guarantees the convergence of the SGD step. Together with the convergence of the optimal transport component, this establishes the overall convergence of the asymptotic DScO-SGD algorithm for RNNs on cognitive tasks.

Before concluding this subsection, we need to point out that the computational costs of the proposed algorithms are dominated by matrix multiplication, rather than the sorting step. Specifically, the computational cost consists of two components:

- SGD step: standard BPTT, with complexity $O(N^2T)$ per trial (same as any RNN training), where T is the number of time steps;
- Optimal transport step: dominated by sorting ($O(N \log N)$ for each of J^E and J^I) and MLE ($O(N)$).

In practice, the Optimal transport step adds approximately 5–10% overhead compared to standard SGD training. The total training times for our experiments range from ~ 5 minutes (Flip-flop, $N = 200$) to ~ 70 minutes (MWG, $N = 1000$) on a single GPU (RTX3080ti 12G).

A.4. Training heavy-tailed RNNs for different tasks

In this paper, we evaluated the asymptotic DScO-SGD method on the following well-known experimental paradigms in cognitive neuroscience.

- *Cue-Set-Go task*: A flexible timing task where the network must generate a specific time interval proportional to the magnitude of an input cue (Wang et al., 2018). Success requires the model to learn a continuous mapping between the static cue amplitude and the target output duration.
- *Measure-Wait-Go task*: A temporal reproduction task initiated by two input pulses (Beiran et al., 2023). The network is tasked with measuring the interval between these pulses, maintaining this duration in working memory across a variable delay, and finally reproducing it upon receiving a ‘Set’ signal.
- *Flip-flop task*: A robust working memory benchmark (Sussillo & Barak, 2013) requiring the network to function as a bistable system. It receives transient pulses via two independent channels and must persistently encode the sign (positive or negative) of the most recent pulse for each channel.

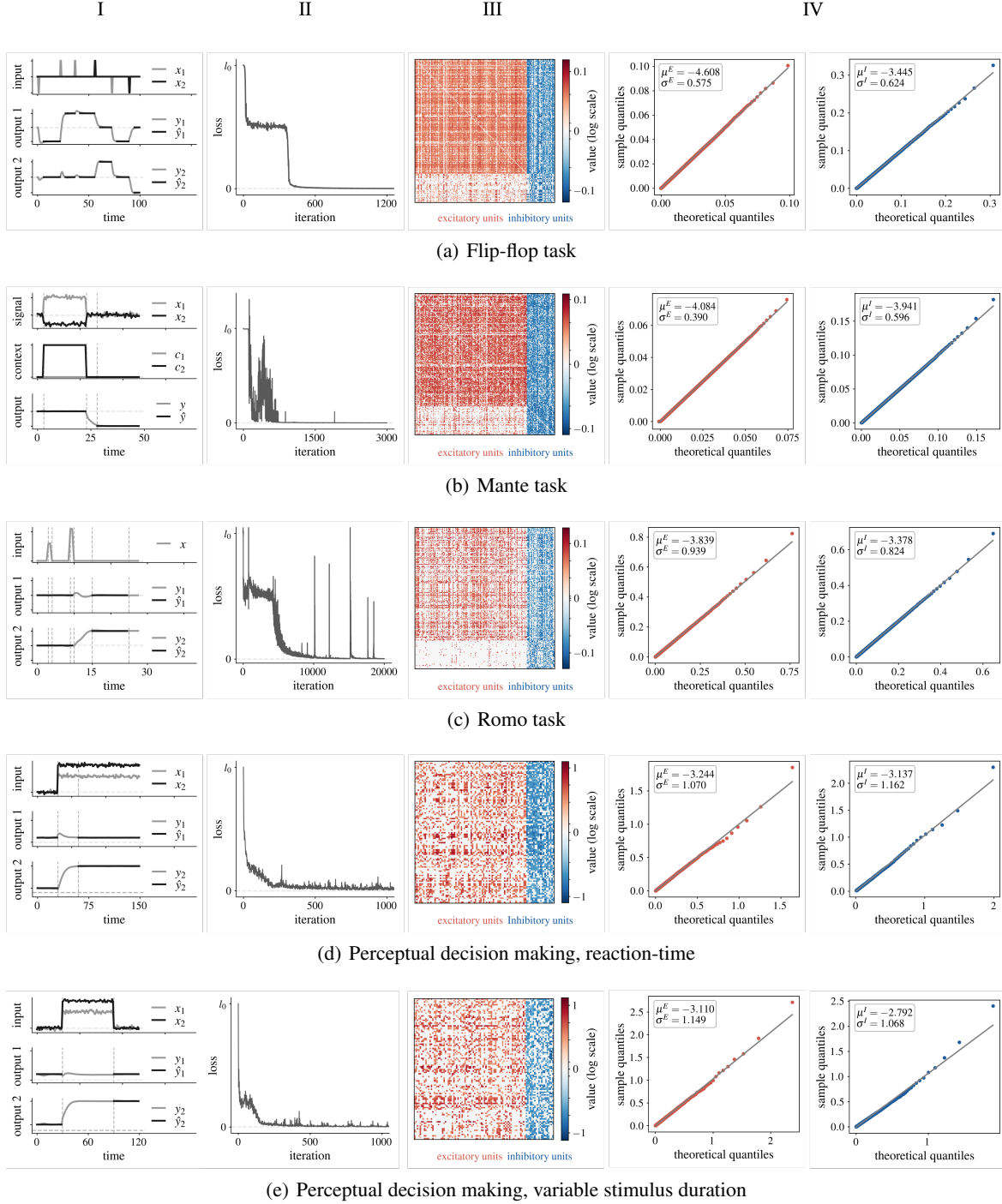


Figure 9. Training heavy-tailed RNNs for different neuroscience tasks. (I) Structure of tasks. (II) Loss function with respect to the iteration step. (III) Heat map of neural connectivities. (IV) Quantile-Quantile plot of E-I units.

- *Mante task*: A context-dependent decision-making paradigm (Mante et al., 2013). The network receives two streams of noisy sensory evidence alongside a context signal. It must selectively integrate the relevant stream while ignoring the irrelevant one to determine whether the average input was positive or negative.
- *Romo task*: A parametric working memory and discrimination task (Romo et al., 1999). Two scalar stimuli are presented sequentially, separated by a delay period. The network must compare their amplitudes and report which of the two stimuli was larger.
- *Perceptual decision making*: A fundamental cognitive process involving the categorization of sensory stimuli (Newsome et al., 1989; Kiani et al., 2008). This task requires the network to identify a stimulus and select the correct response from a set of discrete alternatives. This task is typically performed under two paradigms: the variable stimulus duration (VS) setting, where the evidence accumulation time is controlled by the experimenter, and the reaction-time (RT) setting (Roitman & Shadlen, 2002), where the subject freely determines when to respond based on an internal decision threshold.

The parameters and their numerical values used in the simulations can be found in Table 2. The results of applying the training method (Algorithm 1) to these tasks are shown in Fig. 9.

A.5. Proof of Theorem 3.1

Proof. We note that l given in Section 2 is bounded in $[0, b]$ for some finite constant $b > 0$. This boundness follows from three factors: (1) the activation function $\phi(\cdot) = \tanh(\cdot)$ is bounded in $[-1, 1]$; (2) the readout $z_l(t) = \mathbf{w}_l^\top \phi(\mathbf{x}(t))$ is therefore bounded by $\|\mathbf{w}_l\|_1$; and (3) the target outputs $\hat{z}_l^{(q)}(t)$ are bounded by task design (see Table 2: output amplitudes are in $[-0.5, 0.5]$ or $[0.2, 1.0]$). Consequently, $l = \frac{1}{N_{\text{out}}} \sum_l \sum_t |z_l(t) - \hat{z}_l(t)|^2$ is bounded in $[0, b]$ for some finite b depending on $\|\mathbf{w}_l\|_1$ and the output amplitude range.

By Hoeffding’s lemma (e.g. (Massart, 2007)) we have

$$\mathbb{E} \left[e^{\zeta(l - \mathbb{E}[l])} \right] \leq e^{\frac{\zeta^2 b^2}{8}}.$$

Let $s = \frac{b}{2}$. We then obtain that l is s^2 -subgaussian⁶. In the following, we will show that for any s^2 -subgaussian function l , with probability at least $1 - \beta$ over dataset S , we have

$$\mathbb{E}_{\rho_S}[\varepsilon] \leq 2s \sqrt{\frac{\text{KL}(\rho_S || \pi) + \log \frac{3}{\beta}}{N_{\text{tr}}}}.$$

To that end, we employ Theorem 2.1 in (Germain et al., 2009) to the measurable function $\Psi = \frac{N_{\text{tr}}}{4s^2} \varepsilon^2$, which gives that, with probability at least $1 - \beta$ under S

$$\mathbb{E}_{\rho_S}[\Psi] \leq \log \frac{1}{\beta} + \text{KL}(\rho_S || \pi) + \log \mathbb{E}_{\pi} \mathbb{E}_S [e^{\Psi}]. \quad (8)$$

By Tonelli’s theorem (Folland, 1999) we get

$$\mathbb{E}_S \left[e^{\frac{N_{\text{tr}}}{4s^2} \varepsilon^2} \right] = 1 + \sum_{k=1}^{\infty} \frac{\left(\frac{N_{\text{tr}}}{4s^2} \right)^k}{k!} \mathbb{E}_S [\varepsilon^{2k}], \quad (9)$$

Subsequently, the decomposition (9) is proved to be well-defined and finite. Due to the Hoeffding’s inequality for the s^2 -subgaussian function l (cf. Proposition 2.1 in (Wainwright, 2019)), we have

$$\mathbb{P}(\varepsilon^{2k} \geq u) = \mathbb{P}(|\varepsilon| \geq u^{\frac{1}{2k}}) \leq 2 \exp \left\{ -\frac{N_{\text{tr}} u^{\frac{1}{k}}}{2s^2} \right\}.$$

⁶See Definition 2.1 in Section 2.1.2 of (Wainwright, 2019)

Table 2. Task parameters. $\star$ is randomly chosen from [1, 2, 4, 8, 16].

	CSG task	MWG task	Flip-flop task	Mante task	Romo task	VS	RT
Structure	Fig. 1a	Fig. 1e	Fig. 9a-I	Fig. 9b-I	Fig. 9c-I	Fig. 9d-I	Fig. 9e-I
Input	2	3	2	4	1	3	3
Output	1	1	2	1	2	2	2
Input amplitude	$[0, \frac{1}{12}, \frac{2}{12}, \frac{3}{12}]$		1	1	$U(0.5, 1.5)$	$\frac{1+3.2 \times \frac{\star}{100}}{2}$	$\frac{1+3.2 \times \frac{\star}{100}}{2}$
Input interval		[800, 1050, 1300, 1550]	$U(5, 25)$		0.5		
Output amplitude	$[-0.5, 0.5]$	$[-0.5, 0.5]$	0.5	0.5	60	[0.2, 1.0]	[0.2, 1.0]
Trial duration	350	1000	100	96	3	$40 + U(8, 150)$	90
Fixation duration	$200 + U(-10, 10)$	$250 + U(-10, 10)$	1	3	1	10	10
Stimulation duration	350	1	1	20	$U(2, 8)$	$U(8, 150)$	80
Stimulation delay		[80, 105, 130, 155]	$U(5, 25)$		10		
Decision duration	[80, 105, 130, 155]	[80, 105, 130, 155]		20	5	30	50
Decision delay		150	5	5			30
unit time constant	100ms	100ms	1ms	1ms	1ms	100ms	100ms
learning rate η_1	1.0×10^{-3}	2.0×10^{-4}	2.5×10^{-4}	2.5×10^{-4}	2.5×10^{-4}	1.0×10^{-3}	1.0×10^{-1}
learning rate η_2	$0.1 \times \frac{\text{epoch}}{\text{max epoch}}$	$0.1 \times \frac{\text{epoch}}{\text{max epoch}}$	$0.05 \times \frac{\text{epoch}}{\text{max epoch}}$	$0.01 \times \frac{\text{epoch}}{\text{max epoch}}$	$0.01 \times \frac{\text{epoch}}{\text{max epoch}}$	$0.1 \times \frac{\text{epoch}}{\text{max epoch}}$	$0.1 \times \frac{\text{epoch}}{\text{max epoch}}$
time step	10ms	10ms	500ms	500ms	500ms	10ms	10ms
max epoch	120	340	120	200	2000	180	180
max iteration	1800	5100	1800	3000	30000	2700	2700
Number of Units N	1500	1000	200	200	600	200	200

We further obtain

$$\begin{aligned}
 \mathbb{E}_S[\varepsilon^{2k}] &= \int_0^\infty \mathbb{P}(\varepsilon^{2k} \geq u) du \\
 &\leq 2 \int_0^\infty \exp\left\{-\frac{N_{\text{tr}} u^{\frac{1}{k}}}{2s^2}\right\} du \\
 &= 2k \int_0^\infty \exp\left\{-\frac{N_{\text{tr}} v}{2s^2}\right\} v^{k-1} dv \quad \left(v = u^{\frac{1}{k}}\right) \\
 &= 2k \left(\frac{2s^2}{N_{\text{tr}}}\right)^k \int_0^\infty e^{-tv} v^{k-1} dv \quad \left(t = \frac{N_{\text{tr}}}{2s^2}\right) \\
 &= 2 \left(\frac{2s^2}{N_{\text{tr}}}\right)^k k\Gamma(k) \\
 &= 2 \left(\frac{2s^2}{N_{\text{tr}}}\right)^k \Gamma(k+1) = 2 \left(\frac{2s^2}{N_{\text{tr}}}\right)^k k!,
 \end{aligned} \tag{10}$$

where $\Gamma(\cdot)$ is the Gamma function (cf. pp. 58, (Folland, 1999)). By (9) and (10) we have

$$\mathbb{E}_S \left[e^{\frac{N_{\text{tr}}}{4s^2} \varepsilon^2} \right] \leq 1 + 2 \sum_{k=1}^\infty \frac{\left(\frac{N_{\text{tr}}}{4s^2}\right)^k}{k!} \left(\frac{2s^2}{N_{\text{tr}}}\right)^k k! = 1 + 2 \sum_{k=1}^\infty \frac{1}{2^k} = 3. \tag{11}$$

From (8) and (11) it follows that with probability at least $1 - \beta$ under S one have

$$\frac{N_{\text{tr}}}{4s^2} \mathbb{E}_{\rho_S}[\varepsilon^2] \leq \log \frac{1}{\beta} + \text{KL}(\rho_S || \pi) + \log 3,$$

that is,

$$\mathbb{E}_{\rho_S}[\varepsilon^2] \leq 4s^2 \frac{\text{KL}(\rho_S || \pi) + \log \frac{3}{\beta}}{N_{\text{tr}}}. \tag{12}$$

By Jensen's inequality for convex functions and Fubini's theorem (cf. (Folland, 1999)), we get

$$(\mathbb{E}_{\rho_S}[\varepsilon])^2 \leq \mathbb{E}_{\rho_S}[\varepsilon^2]. \tag{13}$$

Combining (12) and (13) immediately yields the desired result. $\square$

A.6. Proof of Lemma 4.1

Proof. The subsequent proof consists of two parts.

Part 1. Let $W = \frac{1}{N} AA^\top$. Its eigenvalues are denoted by $\varrho_1 \geq \varrho_2 \geq \dots \geq \varrho_N$. Note that $\gamma_k = \sqrt{N\varrho_k}$. We define the corresponding empirical spectral density

$$\mu_N = \frac{1}{N} \sum_{i=1}^N \delta_{\varrho_i}.$$

Then Marchenko–Pastur theory (cf. (Marčenko & Pastur, 1967)) shows that

- $\mu_N \rightarrow \mu_{\text{MP}}$ in weak* topology in distribution, where

$$d\mu_{\text{MP}} = \underbrace{\frac{1}{2\pi} \sqrt{\frac{4-x}{x}} \mathbf{1}_{[0,4]}(x) dx}_{\rho_{\text{MP}}(x)}.$$

Since we have

$$\begin{aligned}
 -\sum_{k=1}^N q_k \ln q_k &= -\sum_{k=1}^N \frac{\gamma_k^2}{\sum_{k=1}^N \gamma_k^2} \ln \left(\frac{\gamma_k^2}{\sum_{k=1}^N \gamma_k^2} \right) \\
 &= -\sum_{k=1}^N \frac{\varrho_k}{\sum_{k=1}^N \varrho_k} \ln \left(\frac{\varrho_k}{\sum_{k=1}^N \varrho_k} \right) \\
 &= \ln \left(\sum_{k=1}^N \varrho_k \right) - \frac{\sum_{k=1}^N \varrho_k \ln \varrho_k}{\sum_{k=1}^N \varrho_k}.
 \end{aligned} \tag{14}$$

then plugging the following equations

$$\sum_{k=1}^N \varrho_k = N \int x d\mu_N, \quad \sum_{k=1}^N \varrho_k \ln \varrho_k = N \int x \ln x d\mu_N$$

into (14) gives

$$-\sum_{k=1}^N q_k \ln q_k = \ln N + \ln \left(\int x d\mu_N \right) - \frac{\int x \ln x d\mu_N}{\int x d\mu_N}. \tag{15}$$

We further have

$$\begin{aligned}
 \frac{r_{\text{eff}}(A)}{N} &= \frac{\exp \left\{ -\sum_{k=1}^N q_k \ln q_k \right\}}{N} \\
 &= \exp \left\{ -\sum_{k=1}^N q_k \ln q_k - \ln N \right\} \\
 &= \exp \left\{ \ln \left(\int x d\mu_N \right) - \frac{\int x \ln x d\mu_N}{\int x d\mu_N} \right\}
 \end{aligned} \tag{16}$$

by (15). Due to the fact that functions $h_1(x) = x$ and $h_2(x) = x \ln x$ are continuous and bounded on $[0, 4]$, we obtain

$$\int x d\mu_N \rightarrow \int x d\mu_{\text{MP}} = 1, \quad \int x \ln x d\mu_N \rightarrow \int x \ln x d\mu_{\text{MP}} =: C_{\text{MP}}, \quad \text{almost surely,} \tag{17}$$

as $N \rightarrow \infty$. Moreover, thanks to the strictly convex property of $h_2(x)$, then by Jensen's inequality we get

$$C_{\text{MP}} > \left(\int x d\mu_{\text{MP}} \right) \ln \left(\int x d\mu_{\text{MP}} \right) = 1 \times 0 = 0.$$

Accordingly, it implies from (16) and (17) that

$$\lim_{N \rightarrow \infty} \frac{r_{\text{eff}}(A)}{N} = e^{-C_{\text{MP}}} < 1, \quad \text{almost surely.}$$

Part 2. We recall the classic location ω_k , defined as the solution of the following integral (cf. (Erdős et al., 2012))

$$\int_{\omega_k}^4 \rho_{\text{MP}}(x) dx = \frac{k}{N}.$$

Since the density function $\rho_{\text{MP}}(x)$ is bounded, by mean value theorems for integrals there exists $\vartheta_k \in [\omega_k, \omega_{k+1}]$ such that

$$\frac{1}{N} = \int_{\omega_{k+1}}^k \rho_{\text{MP}}(x) dx = \rho_{\text{MP}}(\vartheta_k)(\omega_k - \omega_{k+1}),$$

which implies

$$\omega_k - \omega_{k+1} = O\left(\frac{1}{N}\right). \quad (18)$$

From the result on the rigidity of eigenvalues in (Erdős et al., 2012) it implies that there exists a positive constant $C > 0$ such that

$$|\varrho_k - \omega_k| \leq \frac{(\ln N)^{C \ln \ln N}}{N}, \quad \text{w.h.p.} \quad (19)$$

Due to the following equation

$$\varrho_k - \varrho_{k+1} = (\varrho_k - \omega_k) - (\varrho_{k+1} - \omega_{k+1}) + (\omega_k - \omega_{k+1}),$$

we obtain the triangle inequality

$$\varrho_k - \varrho_{k+1} \leq |\varrho_k - \omega_k| + |\varrho_{k+1} - \omega_{k+1}| + |\omega_k - \omega_{k+1}|.$$

By (18) and (19) we further have

$$\varrho_k - \varrho_{k+1} \leq O\left(\frac{(\ln N)^{C \ln \ln N}}{N}\right), \quad \text{w.h.p.} \quad (20)$$

We observe that

$$g_k = \frac{\gamma_k - \gamma_{k+1}}{\gamma_k} = \frac{\sqrt{\varrho_k} - \sqrt{\varrho_{k+1}}}{\sqrt{\varrho_k}} = \frac{\varrho_k - \varrho_{k+1}}{\varrho_k(1 + \sqrt{\frac{\varrho_{k+1}}{\varrho_k}})} \leq \frac{\varrho_k - \varrho_{k+1}}{\varrho_k}. \quad (21)$$

In the bulk of the spectrum, by (19) we obtain $\varrho_k \geq O(1)$. Then combining (20) and (21) gives

$$g_k \leq O\left(\frac{(\ln N)^{C \ln \ln N}}{N}\right), \quad \text{w.h.p.}$$

Consequently, we also have

$$\lim_{N \rightarrow \infty} (\max g_k - \min g_k) = 0, \quad \text{w.h.p.}$$

This concludes the proof. $\square$

A.7. Proof of Lemma 4.2

Let us recall the Rényi entropy of order s (cf. (Bromiley et al., 2004)) that is defined as

$$H_s(\{q_k\}_{k=1}^N) = \frac{1}{1-s} \ln \left(\sum_{k=1}^N q_k^s \right).$$

with $s \in (0, \infty)$ and $s \neq 1$. In particular, we have

$$H_1(\{q_k\}_{k=1}^N) = \lim_{s \rightarrow 1} H_s(\{q_k\}_{k=1}^N) = - \sum_{k=1}^N q_k \ln q_k, \quad H_2(\{q_k\}_{k=1}^N) = - \ln \sum_{k=1}^N q_k^2.$$

Proof. The subsequent proof consists of four steps

Step 1. Let N be the dimension of the matrix and $M = N^2$ be the total number of entries. Let $\{X_k\}_{k=1}^M$ denote the magnitudes of the entries $|A_{ij}|$, and let $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(M)}$ denote their order statistics. From Proposition 3.21 in (Resnick, 2008) it follows that there exists a sequence of normalizing constants $a_M \sim M^{1/\alpha}$ such that the point process of normalized limit values

$$\left\{ \frac{X_{(1)}}{a_M}, \frac{X_{(2)}}{a_M}, \dots \right\}$$

converges weakly to a Poisson point process $\{Y_1, Y_2, \dots\}$ on $(0, \infty]$ with intensity measure $\mu(dx) = \alpha x^{-\alpha-1} dx$.

Since the function $h(x_1, x_2) = \frac{x_1}{x_2}$ is continuous almost everywhere with respect to the limit measure, the joint convergence implies the convergence of the ratio

$$\frac{X_{(1)}}{X_{(2)}} = \frac{X_{(1)}/a_M}{X_{(2)}/a_M} \xrightarrow{\text{weakly}} R. \quad (22)$$

with $R = \frac{Y_1}{Y_2}$. By the maximum domain of attraction (cf. Chapter 12.2.2 (Roncalli, 2020)), we obtain that the distribution function of R satisfies

$$\mathbb{P}(R \leq x) = 1 - \frac{1}{x^\alpha}, \quad x \geq 1.$$

Using the Soshnikov's approximation (Soshnikov, 2004), for any given $k \geq 1$ one has

$$\frac{\gamma_k}{\gamma_{k+1}} \xrightarrow{\mathbb{P}} \frac{X_{(k)}}{X_{(k+1)}}, \quad N \rightarrow \infty. \quad (23)$$

Due to $g_1 = \frac{\gamma_1 - \gamma_2}{\gamma_1} = 1 - \frac{1}{\gamma_1/\gamma_2}$, by (22) and (23) we get

$$g_1 \xrightarrow{\text{weakly}} 1 - \frac{1}{R}.$$

Since R is a random variable supported on $(1, \infty)$, the limit variable $1 - \frac{1}{R}$ is supported on $(0, 1)$. This implies that, there is

$$\lim_{N \rightarrow \infty} \mathbb{P}(g_1 > \epsilon) = P(1 - 1/R > \epsilon) > 0.$$

for any $\epsilon > 0$.

Step 2. Let $Z_k = \ln X_k \sim \mathcal{N}(0, \sigma^2)$. Then $X_{(k)} = \exp\{Z_{(k)}\}$. Let $\hat{Z}_k = \frac{Z_k}{\sigma}$, where $\hat{Z}_k \sim \mathcal{N}(0, 1)$. The ratio of consecutive order statistics is:

$$\frac{X_{(k)}}{X_{(k+1)}} = \exp\{Z_{(k)} - Z_{(k+1)}\} = \exp\{\sigma(\hat{Z}_{(k)} - \hat{Z}_{(k+1)})\}. \quad (24)$$

Let $b_M = \sqrt{2 \ln M}$. Then by using Proposition 2.5 in (Resnick, 2008) we get

$$\left\{ kb_M(\hat{Z}_{(k)} - \hat{Z}_{(k+1)}) \right\}_{k=1}^K \xrightarrow{\text{weakly}} (V_1, \dots, V_K), \quad (25)$$

where $\{V_k\}$ are i.i.d. exponential random variables, i.e., $P(V_k \leq x) = 1 - e^{-x}$. Here, K is a positive integer to be determined later.

Next, we are going to seek an index k^* such that the k -th gap $\frac{X_{(k^*)}}{X_{(k^*+1)}} \geq e^{\hat{\alpha}\sigma^2}$. If some k -th gap satisfies this condition, it follows from (24) that

$$\sigma(\hat{Z}_{(k)} - \hat{Z}_{(k+1)}) \geq \hat{\alpha}\sigma^2 \iff kb_M(\hat{Z}_{(k)} - \hat{Z}_{(k+1)}) \geq kb_M\hat{\alpha}\sigma.$$

Let $\mathcal{A}_k$ be the event that the k -th gap satisfies this condition. Then by (25) the probability of failure for a specific k in the limit is

$$\lim_{N \rightarrow \infty} \mathbb{P}(\mathcal{A}_k^c) = \mathbb{P}(V_k \leq kb_M\hat{\alpha}\sigma) = 1 - \exp(-kb_M\hat{\alpha}\sigma),$$

where α is chosen to satisfy $b_M\hat{\alpha} = O(1)$.

Let $\mathcal{F}_K$ be the event that none of the first K gaps satisfy this condition, that is,

$$\mathcal{F}_K = \mathcal{A}_1^c \cap \dots \cap \mathcal{A}_K^c.$$

Using the asymptotic independence of V_k yields

$$\lim_{N \rightarrow \infty} \mathbb{P}(\mathcal{F}_K) = \prod_{k=1}^K (1 - \exp(-kb_M\hat{\alpha}\sigma)).$$

Since each term in the product is strictly less than 1, the product decays exponentially with K . For any $\delta > 0$, we can choose $K = K(\delta, \sigma)$ sufficiently large such that

$$\mathbb{P}(\exists k^* \leq K \text{ s.t. } \mathcal{A}_{k^*} \text{ holds}) = 1 - \mathbb{P}(\mathcal{F}_K) \geq 1 - \delta. \quad (26)$$

Combining (26) and (23) gives

$$\mathbb{P}(\exists k^* \leq K \text{ s.t. } g_{k^*} \geq 1 - e^{-\hat{\alpha}\sigma^2} \text{ holds}) \geq 1 - \delta. \quad (27)$$

Step 3. Assume $|A_{ij}|$ follows a heavy-tailed distribution with tail index $\alpha \in (0, 2)$ (Infinite variance regime). Let $S_N = \sum_{i=1}^M X_i^2$. Then using Proposition 4.20 in (Resnick, 2008) yields

$$\left(\frac{X_{(1)}^2}{S_N}, \frac{X_{(2)}^2}{S_N}, \dots \right) \xrightarrow{\text{weakly}} (W_1, W_2, \dots), \quad (28)$$

where $\{W_k\}_{k \geq 1}$ is a random vector on the infinite simplex $\sum_{k=1}^N W_k = 1$, and $W_1 > 0$ almost surely. Similarly, it follows from the Soshnikov's approximation that

$$q_k \xrightarrow{\mathbb{P}} \frac{X_{(k)}^2}{S_N}. \quad (29)$$

By (28) and (29), we further have

$$H_1(\{q_k\}_{k=1}^N) \xrightarrow{\text{weakly}} H_1(\{W_k\}_{k=1}^\infty) = - \sum_{k=1}^\infty W_k \ln W_k, \quad N \rightarrow \infty.$$

Note that $H_1(\{W_k\}_{k=1}^\infty)$ is a well-defined random variable, and bounded in probability. Therefore, we have

$$\lim_{N \rightarrow \infty} \frac{\exp\{H_1(\{p_k\}_{k=1}^N)\}}{N} = 0.$$

in probability.

Step 4. From (27) it follows that with probability at least $1 - \delta$ one has

$$\left(\frac{\gamma_{k^*+1}}{\gamma_{k^*}} \right)^2 \leq e^{-2\hat{\alpha}\sigma^2}. \quad (30)$$

Let $P_H = \sum_{i=1}^{k^*} q_i$ and $P_T = \sum_{i=k^*+1}^N q_i$. Then we have $P_H + P_T = 1$. By (30) we observe

$$\frac{P_T}{P_H} = \frac{\sum_{i=k^*+1}^N \gamma_i^2}{\sum_{i=1}^{k^*} \gamma_i^2} \leq \frac{(N - k^*)\gamma_{k^*+1}^2}{\gamma_{k^*}^2} \leq (N - k^*)e^{-2\hat{\alpha}\sigma^2},$$

which further implies

$$P_T = \frac{\frac{P_T}{P_H}}{1 + \frac{P_T}{P_H}} \leq (N - k^*)e^{-2\hat{\alpha}\sigma^2}. \quad (31)$$

Setting $\hat{q}_k = \frac{q_k}{P_H}$ for $k = 1, 2, \dots, k^*$ and $\hat{q}_k = \frac{q_k}{P_T}$ for $k = k^* + 1, k^* + 2, \dots, N$. Then we get

$$\sum_{i=1}^{k^*} \hat{q}_i = 1, \quad \sum_{i=k^*+1}^N \hat{q}_i = 1.$$

We further have

$$\begin{aligned}
 H_1(\{q_i\}_{i=1}^N) &= -\sum_{i=1}^{k^*} q_i \ln q_i - \sum_{i=k^*+1}^N q_i \ln q_i \\
 &= -\sum_{i=1}^{k^*} P_H \hat{q}_i \ln(P_H \hat{q}_i) - \sum_{i=k^*+1}^N P_T \hat{q}_i \ln(P_T \hat{q}_i) \\
 &= -\sum_{i=1}^{k^*} P_H \hat{q}_i (\ln P_H + \ln \hat{q}_i) - \sum_{i=k^*+1}^N P_T \hat{q}_i (\ln P_T + \ln \hat{q}_i) \\
 &= -P_H \ln P_H - P_H \sum_{i=1}^{k^*} \hat{q}_i \ln \hat{q}_i - P_T \ln P_T - P_T \sum_{i=k^*+1}^N \hat{q}_i \ln \hat{q}_i \\
 &= -P_H \ln P_H - P_T \ln P_T + P_H H_1(\{\hat{q}_k\}_{k=1}^{k^*}) + P_T H_1(\{\hat{q}_i\}_{i=k^*+1}^N).
 \end{aligned} \tag{32}$$

Combining the first two terms of (32) gives a binary entropy, which satisfies

$$-P_H \ln P_H - P_T \ln P_T \leq \ln 2. \tag{33}$$

We also note the following estimates

$$P_H H_1(\{\hat{q}_k\}_{k=1}^{k^*}) \leq \ln k^*, \quad P_T H_1(\{\hat{q}_i\}_{i=k^*+1}^N) \leq \ln N. \tag{34}$$

From (31), (32), (33) and (34) we deduce that

$$\begin{aligned}
 H_1(\{q_i\}_{i=1}^N) &\leq \ln 2 + \left[1 - (N - k^*)e^{-2\hat{\alpha}\sigma^2}\right] \ln k^* + (N - k^*)e^{-2\hat{\alpha}\sigma^2} \ln N \\
 &\leq \ln 2 + \ln k^* + N e^{-2\hat{\alpha}\sigma^2} \ln N,
 \end{aligned}$$

which implies

$$\exp\{H_1(\{p_k\}_{k=1}^N)\} \leq \exp\left\{\ln 2 + \ln k^* + N e^{-2\hat{\alpha}\sigma^2} \ln N\right\} = 2k^* \exp\left\{N e^{-2\hat{\alpha}\sigma^2} \ln N\right\} \tag{35}$$

Moreover, if we choose $\hat{\alpha} = \frac{C}{\sqrt{\ln N}}$ and $\sigma^2 \gg \frac{1}{2C}(\ln N)^{3/2}$ with C some constant, then one get $N e^{-2\hat{\alpha}\sigma^2} \ln N \ll 1$. This allows us to apply a Taylor expansion, which from (35) implies

$$r_{\text{eff}} = \exp\{H_1(\{p_k\}_{k=1}^N)\} \leq 2k^* + 3k^* N e^{-2\hat{\alpha}\sigma^2} \ln N.$$

Since $\lim_{\sigma \rightarrow \infty} \sigma^2 e^{-(2\hat{\alpha}-c)\sigma^2} = 0$ for $0 < c < 2\hat{\alpha}$, there exists a positive constant K such that $\sigma^2 e^{-2\hat{\alpha}\sigma^2} \leq K e^{-c\sigma^2}$. Then we have the following estimate

$$3k^* N e^{-2\hat{\alpha}\sigma^2} \ln N \leq 3k^* N e^{-2\hat{\alpha}\sigma^2} (2\hat{\alpha}\sigma^2) = (6K\hat{\alpha}k^*) N e^{-c\sigma^2}.$$

This concludes the proof. $\square$

A.8. Proof of Theorem 4.4

Proof. Let $p_k = \frac{\lambda_k}{\sum_{k=1}^N \lambda_k}$. Then we have

$$r_{\text{pr}}(\{C_{ij}\}) = \frac{\left(\sum_{k=1}^N \lambda_k\right)^2}{\sum_{k=1}^N \lambda_k^2} = \frac{1}{\sum_{k=1}^N p_k^2} = \exp\left\{-\ln \sum_{k=1}^N p_k^2\right\}.$$

By using the reduced singular value decomposition (SVD) of J

$$J = \sum_{k=1}^N \gamma_k \mathbf{w}_k \mathbf{v}_k^\top$$

where $\mathbf{w}_k$ ($\mathbf{v}_k$) is called the k -th left (right) singular vector. We note that $\mathbf{w}_k^\top J = \gamma_k \mathbf{v}_k^\top$. Since the solution of (1) can be written as (cf. Chapter 10 in (Chung & Williams, 1990))

$$\mathbf{x}(t) = \mathbf{x}(0) + \frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} [J\varphi(\mathbf{x}(s)) + I\mathbf{u}(s)] ds + \frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} d\mathbf{B}(t),$$

where $\mathbf{x} = (x_1, \dots, x_N)$, $I = \{I_i\}$ is input matrix and $B(t)$ is the N dimensional Wiener process, we then obtain

$$\begin{aligned} \mathbf{w}_k^\top \mathbf{x}(t) &= \mathbf{w}_k^\top \mathbf{x}(0) + \frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} [\mathbf{w}_k^\top J\varphi(\mathbf{x}(s)) + \mathbf{w}_k^\top I\mathbf{u}(s)] ds + \frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} \mathbf{w}_k^\top d\mathbf{B}(t) \\ &= \underbrace{\mathbf{w}_k^\top \mathbf{x}(0) + \frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} \mathbf{w}_k^\top I\mathbf{u}(s) ds}_{I_1} + \underbrace{\frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} [\gamma_k \mathbf{v}_k^\top \varphi(\mathbf{x}(s))] ds}_{I_2} + \underbrace{\frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} \mathbf{w}_k^\top d\mathbf{B}(t)}_{I_3}. \end{aligned} \quad (36)$$

By Itô isometry (see Theorem 2.3 in (Chung & Williams, 1990)), we have

$$\mathbb{E}[|I_3|^2] = \frac{1}{\tau^2} \int_0^t \left(e^{-\frac{t-s}{\tau}} \right)^2 ds \leq C_1. \quad (37)$$

Additionally, we observe that

$$\begin{aligned} \mathbb{E}[|I_2|^2] &= \left\| \frac{1}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} [\gamma_k \mathbf{v}_k^\top \varphi(\mathbf{x}(s))] ds \right\|_{L^2}^2 \\ &\leq \left(\frac{\gamma_k}{\tau} \int_0^t e^{-\frac{t-s}{\tau}} \|\mathbf{v}_k^\top \varphi(\mathbf{x}(s))\|_{L^2} ds \right)^2. \end{aligned} \quad (38)$$

Note that

$$\|\mathbf{v}_k^\top \varphi(\mathbf{x}(s))\|_{L^2} \leq \|\mathbf{v}_k^\top\|_{L^2} \|\varphi(\mathbf{x}(s))\|_{L^2} \leq \|\varphi(\mathbf{x}(s))\|_{L^2} < \sqrt{N}.$$

By (38) we further have

$$\mathbb{E}[|I_2|^2] \leq \gamma_k^2 N. \quad (39)$$

Thanks to

$$\mathbb{E}[|I_1|^2] \leq C_3,$$

then combining (36), (37) and (39) gives

$$\mathbb{E}[|\mathbf{w}_k^\top \mathbf{x}(t)|^2] \leq \gamma_k^2 N + C_4. \quad (40)$$

For any $\epsilon > 0$, we define the least integer K_ϵ such that

$$\frac{\sum_{k=K_\epsilon+1}^N \gamma_k^2}{\sum_{k=1}^N \gamma_k^2} < \epsilon. \quad (41)$$

Here, we can choose $1 \leq K_\epsilon \leq \exp\left\{\frac{H_1(\{q_k\}_{k=1}^N) + \ln 2}{1-\epsilon}\right\}$ with $q_k = \frac{\gamma_k^2}{\sum_{k=1}^N \gamma_k^2}$. Due to $r_{\text{eff}}(J) = o(N)$, we have

$$\lim_{N \rightarrow \infty} \frac{K_\epsilon}{N} = 0. \quad (42)$$

By using Ky Fan Trace Maximization Principle (cf. (Horn & Johnson, 2012), pp. 248) we obtain

$$\begin{aligned}
 \sum_{k=K_\epsilon+1}^N \lambda_k &= \min \sum_{k=1}^{N-K_\epsilon} \mathbb{E}[|\mathbf{w}_k^\top \varphi(\mathbf{x})|^2] \\
 &\leq C_5 \sum_{k=K_\epsilon+1}^N \mathbb{E}[|\mathbf{w}_k^\top \mathbf{x}(t)|^2] \\
 &\leq C_5 \sum_{k=K_\epsilon+1}^N (\gamma_k^2 N + C_4) \\
 &\leq C_6(N) \sum_{k=K_\epsilon+1}^N \gamma_k^2 \quad (N \text{ large enough}) \\
 &\leq \epsilon C_6(N) \sum_{k=1}^N \gamma_k^2. \quad (\text{by (41)})
 \end{aligned} \tag{43}$$

By Cauchy-Schwartz inequality, we get

$$\left(\sum_{k=1}^{K_\epsilon} p_k \right)^2 \leq K_\epsilon \left(\sum_{k=1}^{K_\epsilon} p_k^2 \right).$$

From (43) it follows

$$\sum_{k=1}^{K_\epsilon} p_k \geq 1 - \epsilon C_7(N)$$

Due to the following estimate

$$\sum_{k=1}^N p_k^2 = \sum_{k=1}^{K_\epsilon} p_k^2 + \sum_{k=K_\epsilon+1}^N p_k^2 \geq \sum_{k=1}^{K_\epsilon} p_k^2 \geq \frac{[1 - \epsilon C_7(N)]^2}{K_\epsilon},$$

we then get

$$\frac{r_{\text{pr}}(\{C_{ij}\})}{N} \leq \frac{K_\epsilon}{N [1 - \epsilon C_7(N)]^2}.$$

By (42) we deduce that $\lim_{N \rightarrow \infty} \frac{r_{\text{pr}}}{N} = 0$. This concludes the proof. $\square$

A.9. Effective ranks for heavy-tailed RNNs on other tasks

Table 3. Effective ranks of heavy-tailed RNNs on different tasks.

	Flip-flop task	Mante task	Romo task	VS	RT
J	Fig. 9a-III	Fig. 9b-III	Fig. 9c-III	Fig. 9d-III	Fig. 9e-III
N	200	200	600	200	200
r_{eff}	7	12	30	10	12
$\frac{r_{\text{eff}}}{N}$	0.0347	0.0602	0.0502	0.0489	0.0605

A.10. Codes

Codes are available at

<https://github.com/zhezhejiao/BioRNN>.

A.11. Figures

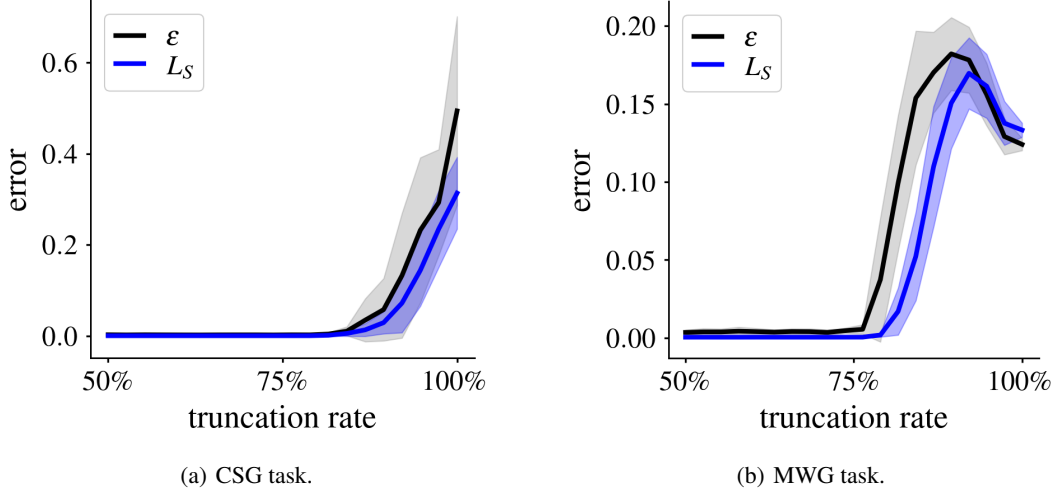


Figure 10. Perturbation test to show the relationship between the truncation rate and errors of training and generalization. We iteratively prune the connection weights, starting from the median and proceeding in ascending order.

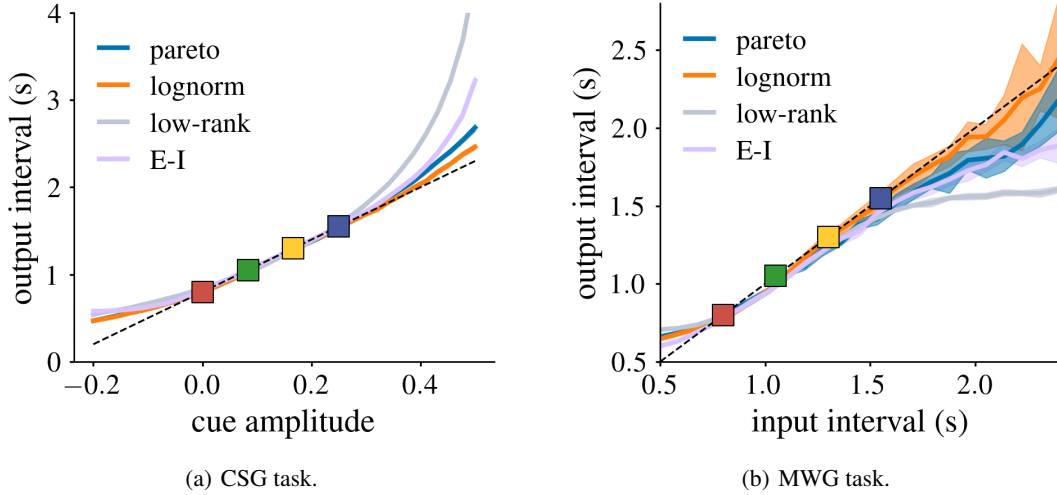
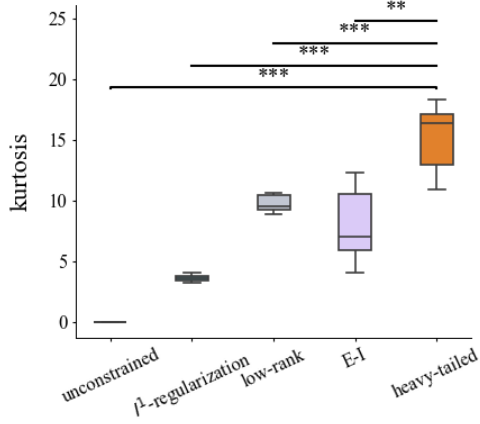
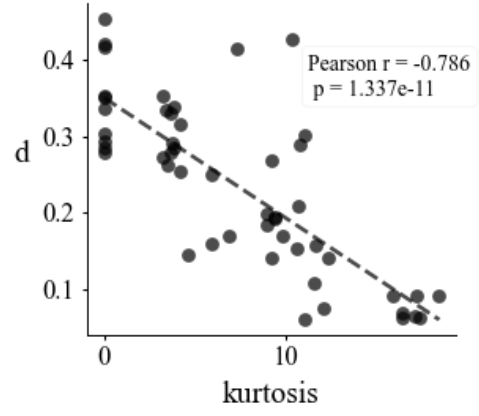


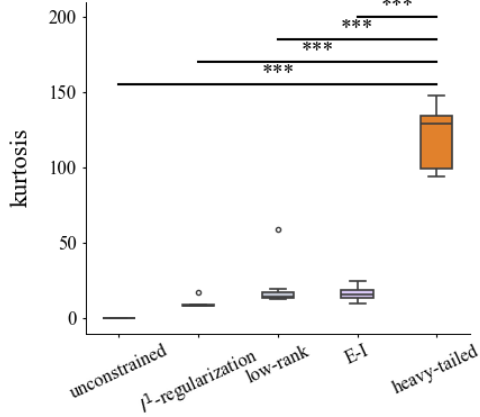
Figure 11. Generalization performance of the trained RNNs with the Pareto-distributed connection.



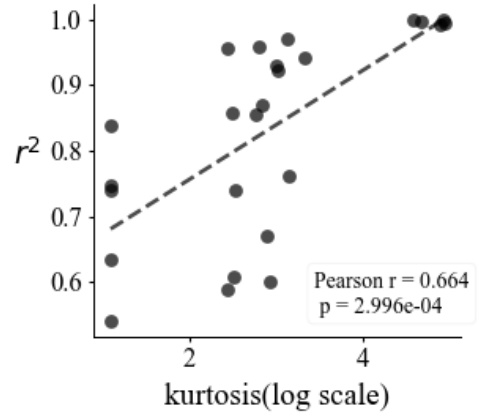
(a)



(b)



(c)



(d)

Figure 12. (a) Kurtosis of different models for the CSG task. (b) Generalization performance versus the excess kurtosis of distributions for the CSG task. (c) Kurtosis of different models for the MWG task. (d) Measurement r^2 of linear representation versus the excess kurtosis for the MWG task.

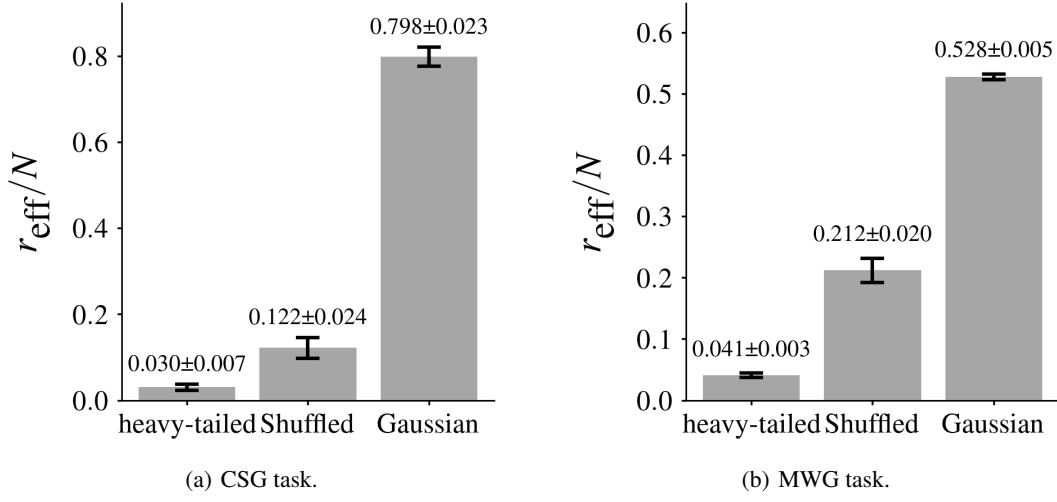


Figure 13. Comparison of the effective ranks across different weight matrices. ‘Heavy-tailed’ refers to the original weight matrix of the heavy-tailed RNN. ‘Shuffled’ denotes the heavy-tailed matrix with its elements randomly permuted. ‘Gaussian’ represents a random matrix whose elements are drawn from a Gaussian distribution with the same mean and variance as the heavy-tailed matrix.

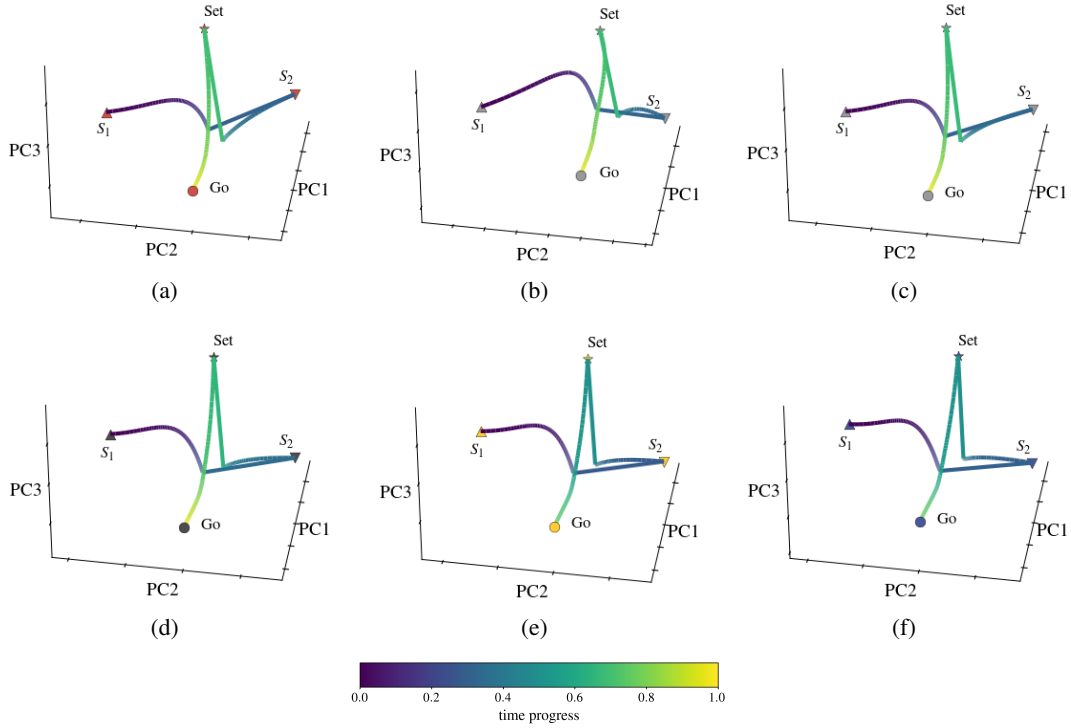


Figure 14. Separate plots of low-dimensional dynamics for MWG task in (Fig. 4b, left).